



Deciding on scheduling, secrecy, and patenting during the new product development process: The relevance of project planning models[☆]

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ABSTRACT

We develop project planning models to integrate scheduling and appropriation decisions for new product development projects. In addition to the decisions related to the timing of development tasks, we focus on whether and when to use secrecy and patenting. We model the setting as a Markov decision process and develop a state-of-the-art dynamic program that enables companies to arrive at an optimal policy for realistically sized instances. By combining an exact analytical characterization of an optimal policy for serial and parallel project networks with extensive numerical experiments for general project networks, we infer that the innovator can benefit from using both secrecy and patenting during different phases of the development project. This contrasts with the traditional view that secrecy and patenting are seen as mutually exclusive appropriation methods. Our results also indicate that opting for secrecy can prolong the project's lead time, and it might therefore conflict with lead-time advantages. Somewhat surprisingly, we find that it might well be optimal to initiate first (rather than to postpone) those tasks that are most likely to expose the project. Combined, these insights highlight the relevancy of an analytical approach that considers decisions related to secrecy, patenting, and the timing of development tasks simultaneously.

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1. Introduction

To protect their intellectual property, firms have to make a choice between a multitude of different appropriation methods. The intensifying pressure for short lead times, the increasing product complexity, and the resulting relevance of collaboration have made the exclusive reliance on single protection mechanisms, such as patents or secrecy, less straightforward [1–3]. Instead, firms need to develop a tailored strategy and to plan carefully what appropriation methods to employ during the different phases of their research and development process.

Examples from diverse industries illustrate that numerous successful firms rely on such customized intellectual property strategies. In consumer electronics, for instance, Apple managed to keep its first iPhone secret during 30 months of development to protect

it from imitation. To keep partners such as Cingular, Yahoo, and Google in the dark as much as possible, the company even developed false prototypes and disguised parts of its software [4]. For the pharmaceutical industry, [5] find empirical evidence that firms time their patent application strategically to postpone the so-called patent cliff – that is to say, the drop in revenue from patent expiry. Additional examples can be found in the food industry, with Kentucky Fried Chicken, McDonald's, and Coca Cola all relying on a mixture of secrecy and patenting [6,7].

Although innovation surveys consistently indicate that secrecy plays a major role in firms' intellectual property strategy, there has been surprisingly little research on how to combine secrecy with other appropriation methods [3]. Instead, there is an almost implicit consensus in the innovation management literature that secrecy and patenting are mutually exclusive substitutes due to the disclosure obligations that pertain to patents [3,8].

In this paper, we examine whether and how the innovator can benefit from a hybrid secrecy/patenting strategy that relies on a mixture of both protection mechanisms during different phases of the new product development (NPD) process. We approach the NPD process as a project, and we use concepts from project man-

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agement and planning to define a fine-grained optimization model that aims to maximize the expected net present value (eNPV). This model simultaneously decides on the sequence in which to perform the project's activities and when to use secrecy or patenting in the project. Extending the work of [9] by incorporating considerations related to secrecy and patenting, we formulate the problem as a Markov decision process and opt for state-of-the-art dynamic-programming techniques to determine an optimal strategy. Finally, we derive managerial insights through an analytical characterization of an optimal policy for serial and parallel project networks, and we corroborate these insights for general project networks using numerical experiments.

Our three main findings are as follows. Firstly, relying on patenting or secrecy as the unique protection mechanism is only desirable in extreme cases where patents provide either very strong or weak protection. For all other situations, the innovator benefits from keeping the innovation secret during the early phases of the project, and from patenting the innovation at a later phase. Hence, our model suggests that secrecy and patenting can be complementary. A hybrid secrecy/patenting strategy is especially beneficial if secrecy becomes increasingly difficult to maintain as the project progresses. Secondly, we find that, although secrecy reduces the time between when competitors learn about the project's existence and product launch, it sometimes increases the project's lead time. Secrecy is therefore not always compatible with lead-time advantages as an appropriation method. Finally, somewhat surprisingly, we find that it might be optimal to initiate first (rather than to postpone) those tasks that are most likely to expose the project. By doing so, the company can test whether they remain undiscovered and, if this is not the case, reorient the project without having incurred the cost of other activities. This finding conflicts with the intuition that it is better to postpone suspicious tasks to delay project discovery [see also 10,11].

To the best of our knowledge, we are the first to apply project scheduling tools, and in particular stochastic dynamic programming, to model decision making related to both secrecy and patenting throughout the NPD process. One important advantage of these tools, which are otherwise typically used to address more operational issues, is that their increased level of detail allows us to cope with the complexity and uncertainty that is characteristic to new product development projects. They enable us to incorporate the timing of secrecy and patenting, as well as to study how different strategies affect the project's lead time. Our results indicate that adopting such a level of detail can indeed benefit companies, since an optimal strategy results from a complex interplay between the different input parameters.

This paper is structured as follows. Section 2 provides a concise review of the literature on secrecy in innovation and project management, and, in Section 3, we formally define our model. Section 4 describes our stochastic dynamic program to solve the problem numerically, and we derive managerial insights in Section 5. Section 6 concludes by indicating how our model can be generalized and by providing possible avenues for further research. All proofs appear in the appendices to this paper.

2. Secrecy in innovation and project management

In the rich innovation management literature, the use of secrecy as a method to protect an innovation from imitation has been widely studied (Section 2.1). Considerably less work, however, has been done with respect to how secrecy can be complementary to other appropriation methods (Section 2.2). There has also been a recent interest in the project management literature in how, from an operational point of view, secrecy should be managed in a project (Section 2.3).

2.1. Secrecy as an appropriation method

Traditionally, patents and trade secrets are conceived as two mutually exclusive means for a company to appropriate the benefits of an innovation. Patents grant a temporary monopoly period (typically 20 years) and, in exchange, require disclosure of the innovation's content. For certain innovations (e.g., the Coca-Cola recipe), however, such disclosure would enable competitors to easily imitate the innovation after patent expiry, or even before if one could invent around certain claims contained in the patent (without jeopardizing the ultimate functionality). That is to say, by using an alternative design or composition, the essential part of an innovation can sometimes be imitated without infringing the patent. Many companies, therefore, refrain from patenting and, instead, rely on informal mechanisms, such as lead-time advantages or secrecy [3]. An additional advantage of trade secrets is that they are not necessarily limited in time.

Initiated by [8], numerous game-theoretic models have been developed in the economics literature to capture different aspects of the trade-off between patent protection and information disclosure. These models reveal that firms are more likely to rely on secrecy rather than patents if the innovation has a high degree of complexity, if there is strong competition, and if the innovation concerns a process rather than a product [3].

Empirical evidence that secrecy frequently plays an important role in firms' appropriation strategies is provided by a series of firm-level surveys [12–16]. These studies consistently indicate that companies use secrecy, patents, and lead-time advantages to protect their innovation from imitation. With respect to these methods' relative importance, secrecy and lead-time advantages are generally found to be more important than patents.

Although this relative comparison suggests that the different appropriation methods are substitutes, [12–14], and [16] all report a positive correlation between firms' ratings for the three appropriation methods. Some scholars have therefore argued that these methods may actually be complementary. Secrecy during the development phase, for example, can reduce the time between when competitors learn about the innovation's existence and the product's launch [14].

2.2. Complementarity between appropriation methods

Concerning the complementarity between secrecy and patents, Arora [17] points out that firms can benefit from patenting the “codifiable” part of their innovation and from keeping secret certain tacit parts, such as know-how. Graham [18] finds empirical support for the hypothesis that firms use continuing patent applications (so-called continuations in part). That is, they consecutively file additional claims on an existing patent to better control the timing of disclosure. For innovations with a significant development time, secrecy also allows firms to file for a patent at a later stage of the project and, in this way, secrecy can extend the effective patent life [19].

This complementarity of secrecy and other types of appropriation fits with the “total intellectual property strategies” as advocated by Granstrand [6]. The idea is to deal with intellectual property in a comprehensive way and to harmonize the different appropriation methods such that the combined effect yields more value. In line with this reasoning, [20] construct a framework that guides firms in their choice between different patenting strategies depending on firm and competitor characteristics. The latter framework, however, does not include the option to combine secrecy and patenting.

Despite the question's relevance, only a few theoretical models examine whether and how innovating companies can use secrecy in tandem with patents to protect the benefits of their innovation

[3]. Those articles that do consider a combined secrecy/patenting strategy focus on subdividing an innovation into parts to patent and parts to keep secret [7,21,22]. Hall et al. [3] mention in this regard that “[e]specially the possibility to rely on patents and secrecy at different stages of the research and development process of an innovation might offer an interesting avenue for further research.” To the best of our knowledge, we are the first to model the joint management of secrecy and patenting during different stages of the research and development process.

2.3. Relevant work from the literature on project planning

The formal mathematical study of project management originated with the Critical Path Management (CPM) method and the Program Evaluation and Review Technique (PERT), which were originally developed to manage the Manhattan Project during World War II [23–25]. Although a project can be defined in multiple ways, most definitions emphasize that it is a unique (i.e., one-off) endeavor in which a set of activities need to be coordinated in order to attain a specified objective [26]. Thus, the development of a new product can be viewed as a project.

Pinker et al. [10, 11] were the first to study how to manage secrecy in the context of a project. Using a nuclear weapon's development project as their main application, the authors examine how the project manager can use a combination of task scheduling, deception, and crashing in order to minimize the exposed time – that is to say, the time between project discovery and completion. The main difference with our model is that we specifically focus on NPD projects rather than military applications. Contrary to [10], our model incorporates uncertainty, the possibility of patenting, and the financial objective of maximizing the project's eNPV. Brown et al. [27] and Gutin et al. [28] study a related problem in which one agent attempts to hinder another agent's project by delaying its activities. Hall [29] also highlights secret project management as a novel area offering numerous interesting research questions.

The problem of maximizing the net present value (NPV) of a project was first studied by Russell [30]. Grinold [31] describes a linear program to solve this problem when all activity durations are deterministic. A more general overview of deterministic project scheduling, including problems with NPV objective and resource constraints, is provided by Kolisch and Padman [32]. The problem of maximizing the project's eNPV when activity durations are exponentially distributed is studied in a sequence of articles [9,33–35], which are all based on the seminal work of [36]. Wiesemann and Kuhn [37] provide an excellent and detailed review of the literature on maximizing a project's eNPV, also for the case where task durations are not exponentially distributed. Our stochastic dynamic program, to be described in Section 4, extends the aforementioned articles by incorporating considerations of secrecy and patenting into the model. Alternative approaches to modeling duration uncertainty in project planning have been studied by, among others, [38] and their references from a more theoretical perspective, and by Kerkhove and Vanhoucke [39] in a more practical context.

3. Model description

In this section, we construct an optimization model that decides on the timing of development activities as well as whether and when to use secrecy and patenting. Figure 1 provides a schematic overview of this model. The focus is primarily on the innovation's pre-market-launch development phase, and the objective is to maximize the innovation's eNPV. The input parameters, to be defined more formally in Section 3.1, are determined by the project characteristics, the relative strength of protection provided

by patenting and secrecy, and the market characteristics. By representing the NPD process as a project, we obtain a fine-grained view that allows us to quantify how the complex interplay between the input parameters and the followed strategy affects the eNPV. Section 3.2 defines this interplay more formally and provides a precise problem statement.

Before formally defining the input parameters of our model in Section 3.1, we first explain how they relate to the different characteristics listed in Fig. 1. The project characteristics determine the activities that need to be performed in order to develop the new product, their intertemporal dependencies in the form of precedence constraints, and their duration and cost. An example of precedence constraints in the context of a new drug development project would be that the Phase II tests could only start after the Phase I tests have been successfully completed. An important assumption that we make is that the duration and cash flow of an activity may depend on whether the task is performed in secrecy or not. This allows us to infer that secrecy may require additional efforts, which might negatively affect the cost and duration and, hence, the overall efficiency of the development tasks. The usage of false prototypes by Apple during the development of its first iPhone illustrates that secrecy can indeed introduce additional costs.

Referring to appropriation (the second column in Fig. 1), a patenting system offers a strong protection if the patent life is long and if it is hard to circumvent a patent by inventing around. Secrecy, in turn, offers a strong protection if the activities are unlikely to generate unintended knowledge spillovers. Our model assumes that the strength of protection provided by secrecy may differ throughout different phases of the project. When developing a new drug, for example, secrecy might be relatively easy to maintain during the early phases of the project, whereas patents may provide better protection as soon as the project enters the testing phase (e.g., because information on the drug's compounds needs to be conveyed to regulatory agencies).

With respect to the market characteristics, the end-of-project cash flow and the expected product life measure the product's market potential. The level of competition in the market, in turn, determines the proportion of the value that remains if the innovator needs to share the market with its competitors because imitation or circumvention occurs. Finally, the discount rate reflects the time-criticality by incorporating the time value of money. We examine one development project in isolation, and we do not look into successive innovation decisions, concerning the timing of different generations of products in a setting of incremental innovation (contrary to Hao et al. [40]).

3.1. Notation and assumptions

Let $N := \{1, \dots, n\}$ collect the n activities that constitute the new product development project, with the final task n reflecting product launch. All other activities $i \in N \setminus \{n\}$ will be referred to as development tasks. The strict partial order $A \subset N \times N$ represents precedence constraints, where activity $j \in N$ can start if and only if all its predecessors $\{i \in N : (i, j) \in A\}$ are completed. We assume that $(i, n) \in A$ for all $i \in N \setminus \{n\}$ such that product launch can only occur if all development tasks have been completed.

Each activity has an uncertain duration and leads to a given cash flow throughout its execution. Let $\delta = 0$ indicate that the project is still a secret and $\delta = 1$ that the innovation has been patented. Due to the disclosure obligation of patents, secrecy can no longer be maintained once the innovation has been patented. For given $\delta \in \{0, 1\}$, we assume that the duration of a development task $i \in N \setminus \{n\}$ is exponentially distributed with parameter $\lambda_i^\delta > 0$, such that its expected duration $1/\lambda_i^\delta$ depends on δ . Executing activity $i \in N \setminus \{n\}$ leads to a cash flow $c_i^\delta \leq 0$ per unit time through-

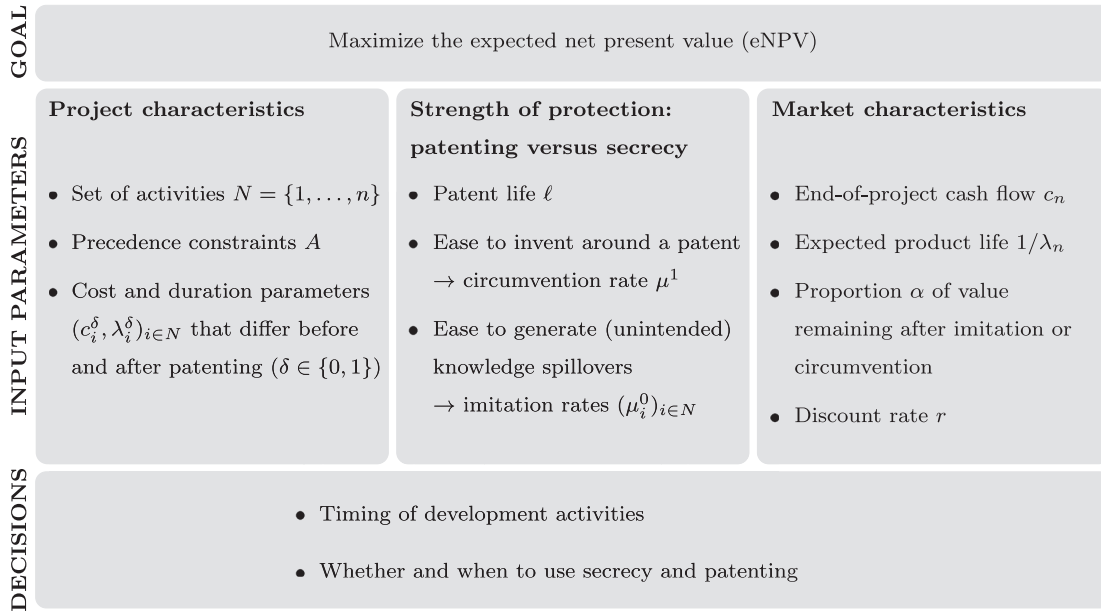


Fig. 1. Schematic overview of the objective, input parameters, and decisions incorporated in the model.

out its duration. Here, the negative cash flow corresponds to a cash outflow, namely, a cost, which also depends on δ . We assume that secrecy does not reduce the activity's expected duration ($\lambda_i^0 \leq \lambda_i^1$) or cost ($|c_i^0| \geq |c_i^1|$). We apply continuous discounting with interest rate $r > 0$ to take into account the time value of money, such that the uninterrupted execution of activity i during the time interval $[0, \tau]$, for fixed $\tau \in \mathbb{R}_{\geq 0}$, yields a net present value

$$\int_0^\tau c_i^\delta e^{-rt} dt.$$

After product launch, the innovator obtains a cash inflow $c_n \geq 0$ per unit time throughout the product's life, whose length we assume to be exponentially distributed with parameter $\lambda_n > 0$. We denote $(\lambda_n^\delta, c_n^\delta) := (\lambda_n, c_n)$ for $\delta \in \{0, 1\}$.

Even if the innovator employs secrecy, the execution of a certain task might still lead to unintended knowledge spillovers that enable competitors to imitate the innovation. By patenting the innovation, the innovator can prevent direct imitation during a patent life $\ell > 0$ (typically 20 years), but during that period they face the risk that the patent will be circumvented (i.e., invented around). Once the patent expires, the innovation is no longer protected and competitors are free to imitate the product. Let $\mu_i^0 \geq 0$ capture the degree to which activity $i \in N$ generates unintended knowledge spillovers, and let the $\mu_i^1 \geq 0$ reflect the ease with which the patent can be circumvented. Given a set of activities $\varphi \subseteq N$ that are being executed, we assume that the remaining time until imitation (if $\delta = 0$) or circumvention (if $\delta = 1$) occurs is exponentially distributed with imitation rate $\sum_{i \in \varphi} \mu_i^0$ or circumvention rate μ^1 , respectively.

If imitation, circumvention, or patent expiry occurs, then the end-of-project cash inflow rate drops to only a fixed share $\alpha \in [0, 1]$ of its original value c_n . This parameter α captures the type and intensity of competition; for example, $\alpha = 0$ in the case of perfect or price-based competition [41]. Let $\delta = 2$ indicate that imitation, circumvention, or patent expiry has occurred, then we define $(\lambda_n^2, c_n^2) := (\lambda_n, \alpha c_n)$ and $(\lambda_i^2, c_i^2) := (\lambda_i^1, c_i^1)$ for all $i \in N \setminus \{n\}$.

Example 1. Consider an example NPD project consisting of five activities with the project network represented in Fig. 2 and the parameters specified in Table 1. The precedence constraints A are the transitive closure of the constraints indicated by the arcs in

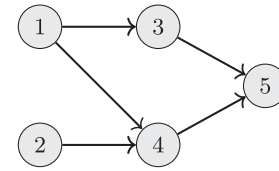


Fig. 2. Example project network in activity-on-the-node (AoN) format.

Table 1

Parameters for the example project.

i	λ_i^0	λ_i^1	c_i^0	c_i^1	μ_i^0
1	0.5	0.5	-9	-9	0.1
2	1	1	-8	-8	1
3	0.1	1	-6	-1	0
4	1	1	-5	-5	0
5	0.2	0.2	50	50	2
$(\mu^1, \alpha, r, \ell) = (0.3, 0.1, 0.01, 10)$					

Fig. 2, so activity 5 has all development tasks $\{1, \dots, 4\}$ as predecessors. Referring to Table 1, the difference for activity 3 in its expected duration ($1/\lambda_3^0 = 10$ vs. $1/\lambda_3^1 = 1$) and cash flow rate ($c_3^0 = -6$ vs. $c_3^1 = -1$) could, for instance, indicate that the innovator has insufficient know-how to perform the task internally and that patenting facilitates collaboration. The non-zero imitation rate of activities 1 and 2 suggests that these tasks possibly generate unintended knowledge spillovers. For example, these activities could correspond to the tasks of performing a large-scale market research in which potential customers are involved. The imitation rate $\mu_n^0 = 2$, in turn, reflects that the product is easily imitated after it is launched (e.g., through reverse engineering). The circumvention rate is taken to be $\mu^1 = 0.3$. If imitation, circumvention, or patent expiry occurs, then there remains only a fraction $\alpha = 0.1$ of the end-of-project cash flow c_n . Finally, the discount rate equals $r = 0.01$ and we assume that one time unit corresponds to half a year, such that with $\ell = 10$ we obtain the typical patent life of 20 years.

To apply our model in practice, the decision maker needs to obtain an estimate for all the parameters introduced above. Given the inherently one-of-a-kind nature of new product development

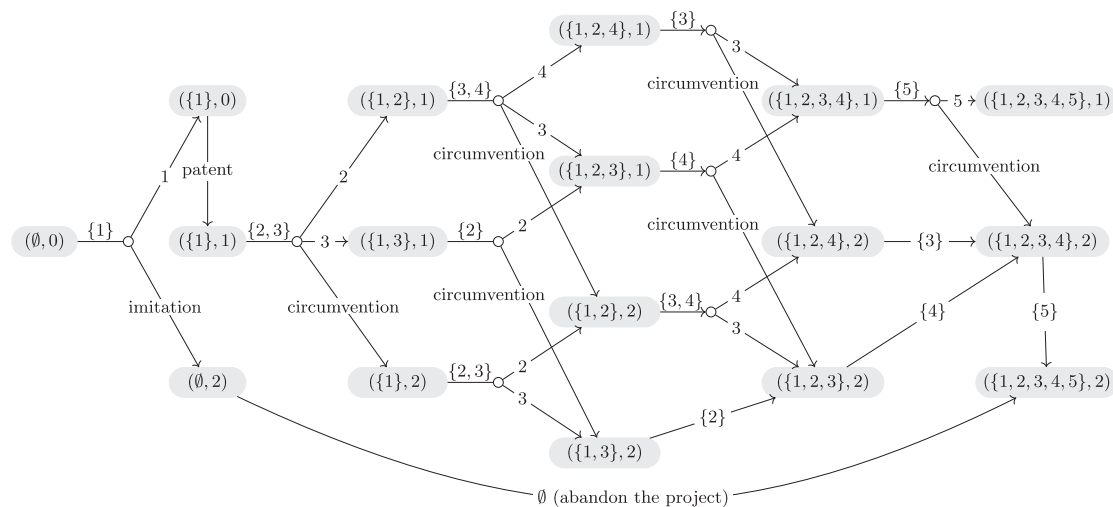


Fig. 3. Decision tree corresponding to an optimal policy for the instance of [Example 1](#). Every node label (P, δ) indicates a decision moment at which the activities $P \subseteq N$ have been completed with corresponding value $\delta \in \{0, 1, 2\}$. Arcs indicate an optimal action (executing a set of activities, patenting the innovation, or abandoning the project) for the corresponding state, possibly followed by a chance node (small circle) to list the possible outcomes that cause a state transition, if this outcome is uncertain. Here, a number on an arc following a chance node reflects that the corresponding activity completes.

projects, obtaining such estimates is not trivial. One way to approach this issue is to first estimate the expected durations and cash flows for a setting without secrecy. To do so, one can take into account the past experience with related development projects and turn to the available methods from the literature [see, for example, 42,43]. Next, one can use expert opinions to categorize the different activities according to whether or not (i) performing them in secrecy would harm their efficiency and (ii) how likely they are to lead to discovery. This categorization then gives insight into how to adjust the baseline parameter estimates so as to consider the effect of operating under secrecy. In order to estimate more precisely the value implications of expiration as well as circumvention—which vary widely across industries—one can rely on a growing literature within the field of innovation/patenting, inspired by foundational work of [44–46], and [47].

3.2. Problem statement

We characterize a solution with a policy that specifies which action to take at which decision moment, possibly contingent on events predating this decision moment [37]. For every set $P \subseteq N$ of activities that have been completed, denote the set of eligible – namely, precedence-feasible – tasks by $E(P) := \{i \in N \setminus P : j \in P \text{ for all } (j, i) \in A\}$. Depending on the eligible activities $E(P)$ at a given decision moment, the feasible actions are then to patent the innovation (if $\delta = 0$) or to execute a set of tasks $\varphi \subseteq E(P)$. Observe that our definition of a feasible action includes the possibility of project abandonment (by choosing $\varphi = \emptyset$) as well as activity preemption (by excluding from φ an uncompleted task that was previously executed).

Let Π be the set of all feasible policies. In combination with the stochastic activity durations and the stochastic time until imitation or circumvention, every policy $\pi \in \Pi$ defines a continuous-time stochastic process $\{\tilde{c}_{it}^{\pi} : t \in \mathbb{R}_{\geq 0}\}$ for each activity $i \in N$, where $\tilde{c}_{it}^{\pi}$ equals the random cash flow generated by activity i at time t . The problem then consists in finding a policy $\pi \in \Pi$ that maximizes the eNPV of the project or, more formally,

$$\max_{\pi \in \Pi} \sum_{i \in N} \mathbb{E} \left[\int_0^{+\infty} \tilde{c}_{it}^{\pi} e^{-rt} dt \right].$$

Example 2. Figure 3 visualizes an optimal policy for the instance of Example 1. The innovator starts by executing activity 1 and

postpones patenting until the task's completion because secrecy currently provides better protection than a patent ($\mu_0^1 < \mu^1$). If imitation occurs before activity 1 completes (leading to node $(\emptyset, 2)$ in the figure), then it is optimal to abandon the project because the reduced end-of-project payoff αc_n does not justify incurring the remaining development costs. Although activity 2 is eligible at the start of the project, its execution is postponed until the innovation has been patented in order to prevent imitation. This patenting also allows the innovator to benefit from activity 3's decreased duration and cost. Once the innovation has been patented, it is optimal to initiate every task as soon as it becomes eligible and to never abandon the project. Using the algorithm of [Section 4](#), one can verify that this policy based on a hybrid secrecy/patenting strategy is indeed optimal and leads to an eNPV equal to 19.80. It therefore significantly outperforms a policy in which the innovator patents the innovation immediately (leading to an eNPV of 15.74) or in which the innovator never patents the innovation (leading to a loss-making project where immediate abandonment is optimal).

4. A stochastic dynamic program to maximize the eNPV

We now describe a novel stochastic dynamic program (SDP) that incorporates elements of secrecy and patenting into the related SDPs by Buss and Rosenblatt [33], Sobel et al. [34], Kulkaarni and Adlakha [36], and Hermans and Leus [9]. The resulting algorithm based on a backward recursion serves as a toolbox that, given the input parameters, allows us to compute an optimal policy. After having defined the SDP in Section 4.1, we show in Section 4.2 how structural results by Hermans and Leus [9] can be generalized to our setting in order to speed up computation times significantly. Thanks to this speedup, we can obtain an optimal policy for realistically sized instances within an acceptable computation time. Finally, Section 4.3 reports on the computational performance of our algorithms. The numerical experiments used to evaluate this performance will also serve to substantiate our analytical insights in Section 5.

4.1. Description of the stochastic dynamic program

Define a state $\sigma = (P, \delta)$ by the set $P \subseteq N$ of finished (past) activities and the indicator $\delta \in \{0, 1, 2\}$ for whether the innovation is

secret ($\delta = 0$), patented ($\delta = 1$), or imitated/circumvented ($\delta = 2$). Let $\mathcal{P}$ collect all sets $P \subseteq N$ such that $i \in P$ implies $j \in P$ for all $(j, i) \in A$, then the state space $\mathcal{S} = \mathcal{P} \times \{0, 1, 2\}$ collects all feasible states. Here, (N, δ) forms a terminal state for each $\delta \in \{0, 1, 2\}$.

Let $\sigma = (P, \delta)$ be a given non-terminal state. As discussed in Section 3.2, the possible actions are then to patent the innovation (if $\delta = 0$) or to execute tasks $\varphi \subseteq E(P)$, where choosing $\varphi = \emptyset$ corresponds to abandonment. If the innovator patents the innovation, then this leads to an instantaneous state transition to state $(P, 1)$. Abandoning the project, in turn, causes a transition to the terminal state (N, δ) . These possible actions and the resulting state transitions are also illustrated in Fig. 3. If the innovator takes an action $\varphi \subseteq E(P)$ with $\varphi \neq \emptyset$, then the next state is determined by the minimum of the time at which the first task $i \in \varphi$ completes, and the time at which imitation or circumvention occurs. By slight abuse of notation, let $\mu^\delta := 0$ if $\delta \in \{0, 2\}$ and, for all $i \in N$, let $\mu_i^\delta := 0$ if $\delta \in \{1, 2\}$. The time until imitation or circumvention occurs is then exponentially distributed with parameter $\mu^\delta(\varphi) := \mu^\delta + \sum_{i \in \varphi} \mu_i^\delta$, such that the transition time is exponentially distributed with parameter $\lambda(\sigma, \varphi) := \mu^\delta(\varphi) + \sum_{i \in \varphi} \lambda_i^\delta$. If activity $i \in \varphi$ completes first, which occurs with probability $\lambda_i^\delta / \lambda(\sigma, \varphi)$, the state becomes $(P \cup \{i\}, \delta)$. If imitation or circumvention occurs first, which occurs with probability $\mu^\delta(\varphi) / \lambda(\sigma, \varphi)$, the state becomes $(P, 2)$. One can verify that the transition time is independent from the event that causes the transition, that the expected discount rate until the next state transition equals $\lambda(\sigma, \varphi) / (r + \lambda(\sigma, \varphi))$, and that the expected cash flow incurred during the state's visit equals $\sum_{i \in \varphi} c_i^\delta / (r + \lambda(\sigma, \varphi))$. For a derivation of these expressions for the transition probabilities and expected cash flows, we refer to [37].

Since it is possible that the product outlives the patent and that no circumvention occurs before patent expiry, we also need to account for the missed profit due to the finite patent life. One difficulty here is that our SDP requires that an optimal action depends on subsequent states only, whereas this missed profit essentially depends on the moment of patenting. To deal with this, we rely on an estimate $\gamma(P)$ for the expected missed profit when patenting in state $(P, 0)$, and charge this cost at the moment of patenting. Appendix B describes how this estimate $\gamma(P)$ is computed. Observe that, via this estimate $\gamma(P)$, we can also easily incorporate the administrative costs associated with patenting into the model. Although these costs can indeed be substantial, we have chosen not to include them so as to limit the number of model parameters.

The considered setting fits within the framework of [48] and, therefore, there is no loss in optimality when choosing actions at state transitions only. We define the value function $V : \mathcal{S} \rightarrow \mathbb{R}$ such that $V(\sigma)$ indicates the maximum eNPV attainable from the remaining project in state $\sigma \in \mathcal{S}$. Given a non-terminal state $(P, \delta) \in \mathcal{S}$, consider an action $\varphi \subseteq E(P)$ with $\varphi \neq \emptyset$, and define $W(P, \delta, \varphi)$ as the eNPV that results when executing tasks φ in state (P, δ) and following an optimal policy afterwards. It then follows from the discussion above that

$$W(P, \delta, \varphi) = \frac{\mu^\delta(\varphi)V(P, 2) + \sum_{i \in \varphi} (\lambda_i^\delta V(P_i, \delta) + c_i^\delta)}{r + \lambda(\sigma, \varphi)}, \quad (1)$$

where we denote $P_i := P \cup \{i\}$ for every $i \in E(P)$. We say that the company abandons the project if it chooses to execute no activities (i.e., $\varphi = \emptyset$), and we define $W(P, \delta, \emptyset) := 0$ to indicate that the remaining value of an abandoned project amounts to zero. For each possible event that can cause a state transition, Eq. (1) combines the probability of this event with the subsequent state's eNPV, discounted to the current state. This is then added to the expected discounted cash flow of taking action φ in the current state.

Table 2

Value and optimal action for each non-terminal state.

State P	Value $V(P, \delta)$			Optimal action			$\gamma(P)$
	$\delta = 0$	$\delta = 1$	$\delta = 2$	$\delta = 0$	$\delta = 1$	$\delta = 2$	
$\emptyset$	19.80	16.88	0.00	{1}	{1, 2}	$\emptyset$	1.14
{1}	42.15	43.00	9.47	patent	{2, 3}	{2, 3}	0.84
{2}	40.77	26.06	0.00	{1}	{1}	$\emptyset$	1.08
{1, 2}	67.74	60.22	17.52	{4}	{3, 4}	{3, 4}	0.73
{1, 3}	47.31	48.11	10.52	patent	{2}	{2}	0.80
{1, 2, 3}	91.59	71.02	18.62	{4}	{4}	{4}	0.66
{1, 2, 4}	73.42	74.08	22.58	patent	{3}	{3}	0.66
{1, 2, 3, 4}	97.50	98.04	23.81	patent	{5}	{5}	0.54

If $\delta = 0$, then the project is still secret and the company can decide to patent the project, leading to an eNPV equal to $V(P, 1) - \gamma(P)$. Here, $\gamma(P)$ equals the estimated missed profit due to the finite patenting life as defined above and as computed in Appendix B. Given the boundary condition $V(N, \delta) = 0$, the value function then satisfies the optimality equation

$$V(P, \delta) = \begin{cases} \max\{\max_{\varphi \subseteq E(P)} W(P, \delta, \varphi), V(P, 1) - \gamma(P)\} & \text{if } \delta = 0; \\ \max_{\varphi \subseteq E(P)} W(P, \delta, \varphi) & \text{otherwise.} \end{cases} \quad (2)$$

Using these equations, we can obtain the maximum eNPV of the project and an optimal policy by computing $V(\emptyset, 0)$ through a backward recursion based on the cardinality of P .

Example 3. For the instance of Example 1, Table 2 displays, for each non-terminal state, the value function, an optimal action, and the estimated missed profit due to the finite patent life. Starting from the initial state $(\emptyset, 0)$, the indicated optimal actions lead to the same optimal policy as reported in Fig. 3. For those states $(P, 0)$ in which patenting is optimal, the difference $V(P, 1) - V(P, 0)$ equals the estimated missed profit due to the finite patent life $\gamma(P)$ as reported in the final column. As soon as all development tasks $i \in \{1, 2, 3, 4\}$ are completed and the innovation has been patented, the company can launch the new product with expected discounted cash inflow $c_5 / (r + \lambda_5 + \mu^1) = 98.04$.

4.2. Structural properties to speed up the algorithm

In this section, we show how the structural properties of the SDP can be exploited in order to improve the running time significantly. As will be demonstrated in the next section, this speedup enables us to determine an optimal policy for realistically sized instances within acceptable computation times. Moreover, the structural properties to be derived will also prove useful in Section 5 to obtain analytical insights.

Theorem 1 below characterizes an optimal policy, and Theorem 2 enables us to compute an optimal action for a given state efficiently. Both theorems, proven in Appendix C–Appendix D, generalize the results obtained in [9] by also including elements of secrecy and patenting.

For each non-terminal state $\sigma = (P, \delta) \in \mathcal{S}$ and eligible task $i \in E(P)$, define

$$\psi_i(\sigma) := \frac{1}{\lambda_i^\delta + \mu_i^\delta} (\lambda_i^\delta V(P_i, \delta) + \mu_i^\delta V(P, 2) + c_i^\delta). \quad (3)$$

To interpret the expression for $\psi_i(\sigma)$, observe that the eNPV in the case of a state transition caused by task i equals $(\lambda_i^\delta V(P_i, \delta) + \mu_i^\delta V(P, 2)) / (\lambda_i^\delta + \mu_i^\delta)$ and that $c_i^\delta / (\lambda_i^\delta + \mu_i^\delta)$ equals the expected cash flow of activity i . Hence, $\psi_i(\sigma)$ forms a measure for the total expected value of executing task i . Theorem 1 states that it is optimal to execute those tasks for which this value $\psi_i(\sigma)$ is larger than or equal to the current state's value $V(\sigma)$.

Theorem 1. Let $\sigma = (P, \delta) \in S$ be a non-terminal state. If it is not optimal to patent the innovation in this state, then executing the set of activities

$$\varphi^*(\sigma) := \{i \in E(P) : \psi_i(\sigma) \geq V(\sigma)\}$$

is an optimal action in this state.

Now define for a non-terminal state $\sigma = (P, \delta) \in S$ and eligible task $i \in E(P)$ the set $\varphi_i(\sigma) := \{j \in E(P) : \psi_j(\sigma) > \psi_i(\sigma)\}$. [Theorem 2](#) states that we should execute task i if the eNPV of its transition, as measured by $\psi_i(\sigma)$, is larger than or equal to the eNPV executing tasks $\varphi_i(\sigma)$ and following an optimal policy afterwards.

Theorem 2. Let $\sigma = (P, \delta) \in S$ be a non-terminal state. If it is not optimal to patent the innovation in this state, then it holds that $\varphi^*(\sigma) = \{i \in E(P) : \psi_i(\sigma) \geq W(P, \delta, \varphi_i(\sigma))\}$.

The above results enable us to improve the running time of our SDP significantly since we can avoid the maximization over all sets $\varphi \subseteq E(P)$ to determine an optimal action in a given state. In particular, let $m := \max_{P \in \mathcal{P}} |E(P)|$ be the maximum number of tasks that can be active in parallel. Consider a given non-terminal state $(P, \delta) \in S$ and assume that the value function is known for all states $(P', \delta') \in S$ with $|P'| = |P| + 1$. [Theorem 2](#) then enables us to evaluate the value function in time $O(m \log(m))$, where the time needed to sort the values $\psi_i(\sigma)$ for all $i \in E(P)$ dominates. This significantly improves the $O(2^m)$ time needed by the more naive implementation as described in [Section 4.1](#). The total running time, however, still grows exponentially in m since we need to determine an optimal action in every state and $|S|$ increases exponentially in m . To see that this is indeed the case, observe that there are at most $\binom{n}{i}$ sets $P \in \mathcal{P}$ with $|E(P)| = i$ for every $i = 0, 1, \dots, m$, such that $|\mathcal{P}| \leq \sum_{i=0}^m \binom{n}{i}$; see also [\[28\]](#). By the binomial theorem, we thus obtain that $(1+n)^m = \sum_{i=0}^m \binom{m}{i} n^i \geq \sum_{i=0}^m n^i \geq \sum_{i=0}^m \binom{n}{i} \geq |\mathcal{P}|$. Hence, we obtain that $|S|$ is $O(n^m)$, such that the total running time of our SDP is $O(m \log(m) n^m)$.

4.3. Computational performance

We now demonstrate that our SDP can solve realistically sized instances to optimality in reasonable computation times. To do so, we use the dataset provided by Creemers et al. [\[35\]](#), which consists of 30 randomly generated instances for each combination of the number of activities $n \in \{11, 21, \dots, 121\}$ and the order strength $OS \in \{0.4, 0.6, 0.8\}$. The order strength measures the project network's density and is defined as the number of precedence constraints $|A|$ divided by the maximum number $n(n-1)/2$ of such constraints. Since [\[35\]](#) do not consider elements of secrecy and patenting, we generated the corresponding additional parameters ourselves; [Appendix A](#) discusses in more detail the dataset of test instances.

[Table 3](#) reports the average CPU time in seconds and the number of solved instances for our SDP with or without the speedup enabled by [Theorems 1–2](#). All experiments were performed with an Intel Core i7-4790 processor with 3.60 GHz CPU speed. We set a memory limit of 2 GB and a time limit of 1 h. The table indicates that the order strength is a major determinant of the computational performance, both for the setting with and without the speedup. Given the theoretical analysis of our SDP's running time in [Section 4](#), this is not surprising since sparser networks (lower OS) allow for more tasks to be executed in parallel (higher m). The table also shows that [Theorems 1–2](#) enable us to solve instances sometimes more than 50 times faster. Since the memory requirements are the same for the SDP with and without the speedup, the one-hour time limit explains why some instances can only be

solved with the speedup. The main bottleneck for the SDP with the speedup, in contrast, is memory rather than CPU time.

[Table 4](#) elaborates on the SDP's memory requirements by reporting the number of states and the maximum fraction of these states kept in memory at any time during the algorithm's execution. Indeed, when evaluating the value function of those states $(P, \delta) \in S$ with $|P| = k$ for some $k = 1, \dots, n-1$, it suffices to have in memory only the value function for the states $(P', \delta) \in S$ with $|P'| = k+1$. Especially for instances with many activities and a high order strength this means that only a relatively small fraction of the states needs to be kept in memory. One consequence of this, however, is that also only a fraction of the computed optimal policy is kept in memory. In practice this means that if, throughout its execution, the project transitions to a state that has been removed from memory, then the decision maker needs to re-execute the SDP to compute an optimal action for that state. Finally, observe that the SDP with and without the speedup have the same state space, so the statistics reported in [Table 4](#) are representative for the memory requirements of both methods.

5. Insights

In this section, we use our model to derive managerial insights by examining the optimal timing of patenting ([Section 5.1](#)) as well as the effect of secrecy on the scheduling decisions ([Section 5.2](#)). We find that, although most models in the literature treat secrecy and patenting as mutually exclusive appropriation methods [\[3\]](#), it is frequently optimal to follow a hybrid strategy that uses both secrecy and patenting in different parts of the project. Moreover, we find that secrecy tends to increase the project's lead time and, therefore, it might conflict with lead-time advantages as a protection mechanism.

To derive these insights, we first determine closed-form expressions for an optimal policy in the specific case of serial and parallel networks (illustrated in [Fig. 4](#)), and subsequently corroborate these analytical findings for more general project networks through numerical experiments. [Sections 5.1](#) and [5.2](#) are structured accordingly: in each subsection, we first describe our analytical findings and subsequently discuss the results of our computational experiments for general project networks.

Serial and parallel project network structures are frequently used in practice to provide a high-level overview for (parts of) the development process. Each activity of a serial project can, for instance, represent a stage in the project, where the detailed scheduling of the activities within a stage is delegated to a lower level of decision making [\[49\]](#). A serial project is also typically used to model so-called stage-gate processes [\[50,51\]](#). A parallel project structure, in turn, reflects a setting in which the activities are unrelated, as is frequently the case for tasks within a specific stage or when working with a modular architecture. Numerous articles in the academic literature also rely on serial and parallel project structures to derive insights; see, for instance, [\[33,52–54\]](#), or [\[28\]](#).

5.1. Optimality of a hybrid secrecy/patenting strategy

Depending on the timing of abandonment and patenting, an optimal policy can be categorized into one of the following four strategies. Either the policy (i) immediately abandons the project, (ii) immediately patents the innovation, (iii) uses secrecy up until product launch, or (iv) relies on a hybrid secrecy/patenting strategy. We say that patenting and secrecy are mutually exclusive for a certain policy if only one of these two methods is used as the unique protection mechanism up until product launch (as is the case for the second and third strategies stated above). For a hybrid secrecy/patenting strategy, in contrast, the innovation is initially kept secret and only patented at a later point in time be-

Table 3

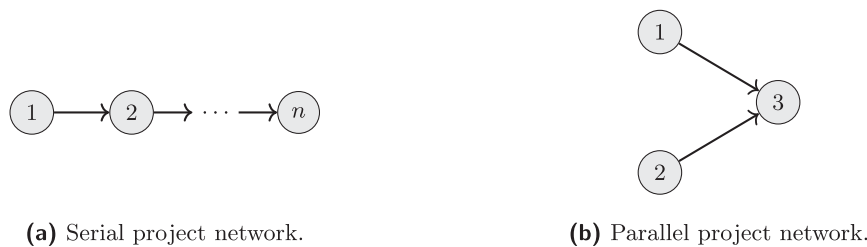
Average CPU time in seconds and number of solved instances (# out of 30 using a memory limit of 2 GB and a time limit of 1 h) for the SDP with or without the speedup enabled by [Theorems 1–2](#).

n	SDP without speedup						SDP with speedup					
	OS=0.80		OS=0.60		OS=0.40		OS=0.80		OS=0.60		OS=0.40	
	CPU	#	CPU	#	CPU	#	CPU	#	CPU	#	CPU	#
11	0.00	30	0.00	30	0.00	30	0.00	30	0.00	30	0.00	30
21	0.00	30	0.00	30	0.03	30	0.00	30	0.00	30	0.01	30
31	0.00	30	0.03	30	0.86	30	0.00	30	0.01	30	0.10	30
41	0.01	30	0.28	30	49.71	30	0.00	30	0.05	30	1.58	30
51	0.01	30	2.33	30	874.34	29	0.01	30	0.23	30	15.50	30
61	0.04	30	23.24	30			0.01	30	1.10	30	164.65	30
71	0.15	30	214.97	30			0.03	30	5.42	30	672.17	20
81	0.39	30	860.35	28			0.06	30	13.35	30		
91	1.28	30	2,719.85	4			0.15	30	75.75	30		
101	3.47	30					0.31	30	286.63	30		
111	12.83	30					0.74	30	850.55	22		
121	81.95	30					2.98	30	3,005.86	2		

Table 4

Average total number of states and average maximum fraction of states in memory at any time for the instances solved using the SDP with speedup with a one-hour time limit and a 2 GB memory limit.

n	OS=0.80		OS=0.60		OS=0.40	
	# states	% in memory	# states	% in memory	# states	% in memory
11	22	29.07	41	32.43	84	33.19
21	88	20.82	330	20.86	1,620	23.8
31	254	14.04	1,898	15.67	17,096	16.82
41	662	12.28	9,480	13.38	193,848	14.43
51	1,544	10.61	40,379	11.89	1,659,710	12.24
61	3,564	9.77	175,288	10.91	13,791,000	11.41
71	7,754	9.64	727,948	10.28	71,386,600	8.04
81	16,364	7.97	2,259,480	8.11		
91	34,057	8.33	9,845,430	8.81		
101	66,670	7.79	31,268,000	8.14		
111	146,910	7.76	85,471,500	6.59		
121	515,785	6.85	207,036,000	3.97		

**Fig. 4.** Specific project network structures.

fore product launch. Observe that, according to the categorization above, a strategy in which secrecy is used during the development phase and the innovation is patented at the moment of product launch falls into the third category. Thus, it is not considered to be a hybrid secrecy/patenting strategy even though it does combine secrecy and patenting. The reason for this distinction is because we wish to focus our model and insights on the innovation's pre-launch development phase. In this section, we show conditions under which a hybrid secrecy/patenting strategy is more likely to be optimal.

Our first proposition characterizes the potential decision moments at which project abandonment should be considered. The intuition behind this result, whose proof appears in [Appendix E](#), is that as long as no imitation or circumvention occurs, the project's remaining value does not decrease throughout its execution. As a result, if the project is profitable at its initiation and no imitation or circumvention occurs, then it remains profitable. Observe that even if imitation or circumvention occurs, then it might still be op-

timal to proceed with the project (and thus to not abandon) if the remaining proportion α of the end-of-project payoff is sufficiently high.

Proposition 1. *There is no loss in optimality by considering abandonment only at the moment of project initiation, or if imitation or circumvention occurs.*

There can be multiple external factors not incorporated in our model that provide an incentive to abandon the project at other decision moments than those listed in [Proposition 1](#). Examples of such incentives include the failure of an activity [\[55,56\]](#) or, if durations are not exponentially distributed, the realization of an exceptionally long duration. The main importance of [Proposition 1](#) is that, within the assumptions of our model, it isolates the incentives to abandon in the absence of such external factors, which in turn will be helpful for the analysis in the remainder of this section.

5.1.1. Analytical approach for serial project networks.

In order to obtain insight into when a hybrid secrecy/patenting strategy is more likely to be optimal, we focus on the specific case of a serial project network with precedence relation $A = \{(i, j) \in N \times N : i < j\}$, as displayed in Fig. 4(a). Moreover, for simplicity, we also assume that, if imitation or circumvention occurs, the end-of-project payoff drops to zero; that is, $\alpha = 0$. We refer to an instance with the above structure as a serial project instance.

Since the precedence relation for a serial project instance forms a total order, the sequence in which the activities need to be executed is fixed. The only decision that remains is therefore whether and, if so, when to patent or abandon the project. Proposition 2, proven in Appendix F, provides sufficient conditions for secrecy and patenting to be mutually exclusive. One can verify that, if an instance violates at least one of the conditions stated in the proposition, then it might only have a hybrid secrecy/patenting strategy as an optimal policy.

Proposition 2. *Consider a serial project instance in which immediate abandonment is not optimal. Secrecy and patenting are then mutually exclusive if the following three conditions are satisfied:*

- (i) $\ell = +\infty$;
- (ii) $(r + \mu^1)/\lambda_i^1 - (r + \mu_i^0)/\lambda_i^0$ is non-decreasing in $i = 1, \dots, n - 1$;
- (iii) $|c_i^0|/\lambda_i^0 - |c_i^1|/\lambda_i^1$ is non-increasing in $i = 1, \dots, n - 1$.

The first condition reflects a setting in which there is no missed profit due to the finite patent life because the project duration and product life are negligibly small compared to the patent life. In fast-moving industries such as consumer electronics, for example, the product life only very rarely exceeds the typical patent life of 20 years. Observe that r , μ^1 , and μ_i^0 reflect the loss in value per time unit due to the interest, circumvention, and imitation rate, respectively. Thus, Condition (ii) states that the degree to which secrecy provides better protection to this loss in value does not decrease as the project progresses. Condition (iii), in turn, states that the difference in a task's expected cost before and after patenting does not increase as the project progresses. Therefore, Conditions (ii)–(iii) reflect a setting in which secrecy does not become more difficult to maintain throughout the project.

In practice, however, it is more frequently the case that secrecy does become increasingly difficult to maintain throughout the project. During the early phases of the project, there are typically fewer people involved, and there is sufficient know-how to perform the activities internally. During the later phases of the project, however, the activities might become less closely related to the innovator's core business. As a result, external knowledge acquisition or collaboration with outside parties (e.g., suppliers and customers) might be required, such that more people get involved in the project. It follows from the models of [57] that it becomes increasingly difficult to maintain secrecy as more people become involved. As such, we find that the innovator might benefit from a hybrid strategy that keeps the innovation secret during the early phases of the project and only patents when secrecy becomes harder to maintain. These dynamics explain why Apple employed a hybrid secrecy/patenting strategy during the development of their first iPhone. When the company had to request permits from the Federal Communications Commission, it became too difficult to keep the secret, and Apple decided to announce their innovation [4].

5.1.2. Numerical experiments for general project networks.

To examine the conditions under which a hybrid secrecy/patenting strategy is more likely to be optimal for general project networks, we performed a series of numerical experiments using the test instances described in Appendix A with $n \in$

$\{11, 21, 31, 41\}$. More specifically, we adapted the baseline instances according to the different experimental settings described in Table A.1, and we subsequently determined an optimal policy for each of these instances.

Figure 5 summarizes the results of our experiments. It clearly shows that, for all considered settings, a hybrid strategy is optimal for a significant proportion of the instances. The setting with an increasing imitation rate corresponds to a situation in which activities with more predecessors (which are thus more likely to occur later in the project) have a higher chance of causing imitation. Similarly, the settings with an increasing cost or duration difference correspond to a situation in which later-occurring activities are more likely to have a higher expected cost and duration difference, respectively. Hence, these three settings capture the case where secrecy becomes increasingly difficult to maintain as the project proceeds. The final setting measures the effect of shortening the patent life, which makes it more likely that the product life exceeds the patent life. Figure 5 shows that, for all these settings, a hybrid strategy is optimal more frequently than in the baseline setting, which is consistent with Proposition 2. We also observe that, for the settings with an increasing cost difference and shorter patent life, the effect is less pronounced than for the other two settings.

5.2. Effect of secrecy on the scheduling decisions

To obtain insight into how secrecy affects the scheduling decisions, we first focus on the parallel project network as displayed in Fig. 4(b) to characterize the structure of an optimal policy analytically. Next, we use numerical experiments to corroborate our insights.

5.2.1. Analytical approach for a parallel project network.

Consider the parallel project with $n = 3$ tasks and $A = \{(1, 3), (2, 3)\}$ as displayed in Fig. 4(b). For each of the two development tasks $i \in \{1, 2\}$ and for each $\delta \in \{0, 1\}$, we consider equal cost and duration parameters $c_i^\delta = c < 0$ and $\lambda_i^\delta = \lambda > 0$. We say that activity 1 is suspicious because it has $\mu_1^0 \geq 0$ and, therefore, it can cause unintended knowledge spillovers that lead to imitation. Activity 2, on the other hand, is said to be innocent because it has $\mu_2^0 = 0$ and thus never causes imitation. We set $\ell = +\infty$ and $\alpha = 0$, and we let $\mu_3^0 = \mu^1$ such that the innovator is indifferent between secrecy and patenting at the moment of project launch. The remaining parameters $(\mu^1, \lambda_3, c_3, r)$ have no additional constraints. We refer to an instance with the above structure as a parallel project instance.

Proposition 3, stated below and proven in Appendix G, shows that, for a parallel project instance, at least one of the policies defined in Table 5 is optimal. To simplify notation, we define the profit potential R of the project as

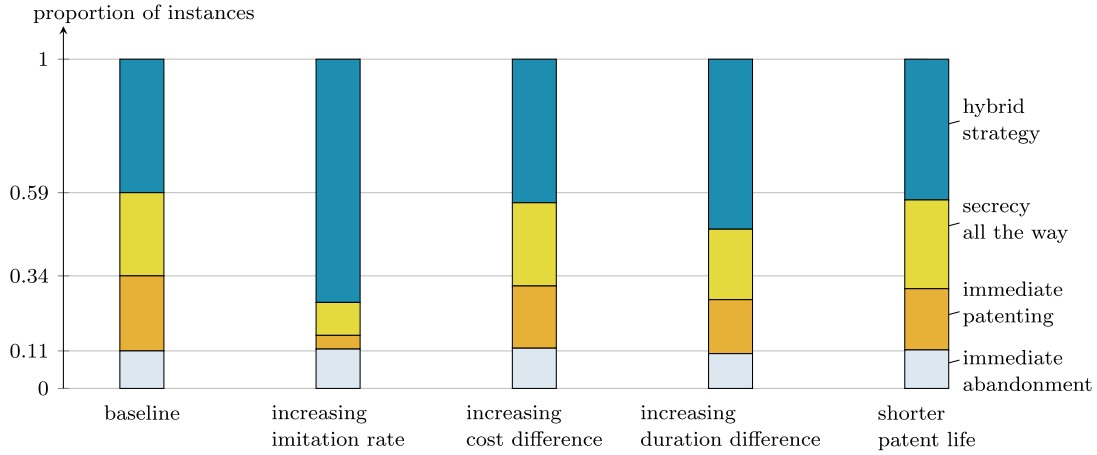
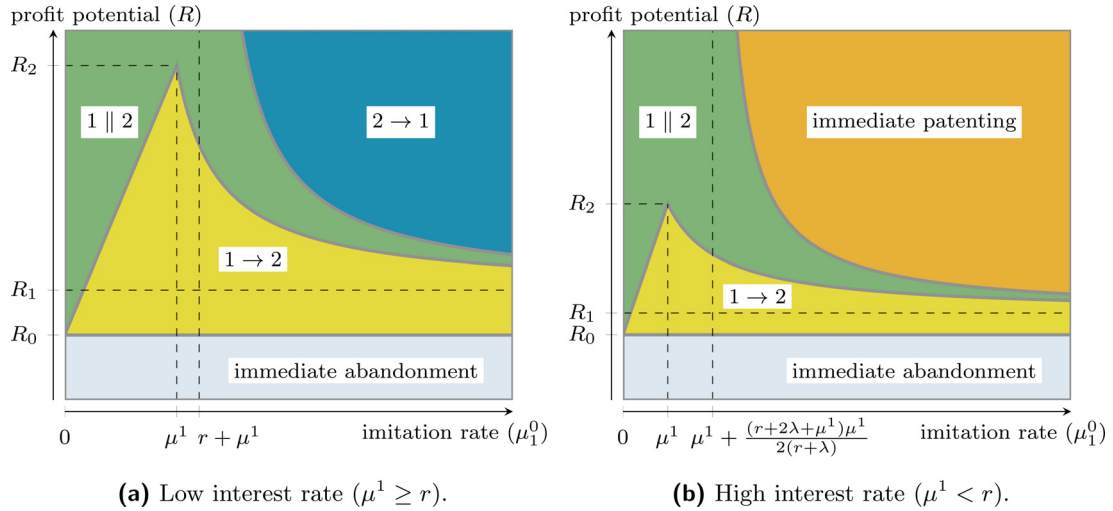
$$R := \left(\frac{\lambda c_3}{r + \lambda_3 + \mu^1} + c \right) \frac{\lambda}{|c|}.$$

Since the expected discounted cash inflow after product launch equals $c_3/(r + \lambda_3 + \mu^1)$, the profit potential R measures the degree to which the end-of-project payoff exceeds the costs associated with the development project. Figure 6 gives a visual representation of Proposition 3 by plotting the optimal policy as a function of the imitation rate μ_1^0 and profit potential R , depending on whether the circumvention rate μ^1 is larger or smaller than the interest rate r . To simplify notation in this figure, we define $R_0 := r + \lambda$, $R_1 := r + \lambda + \mu^1$, and $R_2 := (r + \lambda)(1 + \mu^1/r)$.

Table 5

Description of potentially optimal policies for a parallel project instance. Since $\alpha = 0$ by assumption, imitation or circumvention always leads to abandonment for the described policies.

Policy	Description
Immediate abandonment	Abandon the project immediately
$1 \rightarrow 2$	Perform activity 1 before 2 without patenting
$2 \rightarrow 1$	Perform activity 2 before 1, and patent when initiating activity 1
$1 \parallel 2$	Perform activities 1 and 2 in parallel, and only patent the innovation after activity 2 completes if $\mu_1^0 > \mu^1$
Immediate patenting	Patent the innovation, and perform activities 1 and 2 in parallel

**Fig. 5.** Relative frequency with which the different strategies are optimal for general project networks.**Fig. 6.** Optimal policy as a function of the imitation rate μ_1^0 and profit potential R for parallel project instance in case $\mu^1 \geq r$ (left) or $\mu^1 < r$ (right). Here, $R_0 := r + \lambda$, $R_1 := r + \lambda + \mu^1$, and $R_2 := (r + \lambda)(1 + \mu^1/r)$.

Proposition 3. For a parallel project instance, it holds that:

- (i) immediate abandonment is optimal if and only if $R \leq r + \lambda$;
- (ii) $1 \rightarrow 2$ is optimal if $\mu_1^0 \leq \mu^1$ and $r + \lambda < R < (r + \lambda)(1 + \mu_1^0/r)$, or if $\mu_1^0 \geq \mu^1$ and

$$r + \lambda < R < (r + \lambda + \mu^1) \frac{(r + \lambda)(r + \mu_1^0)}{(r + \lambda)(r + \mu_1^0) - \lambda \mu^1};$$

- (iii) $2 \rightarrow 1$ is optimal if $\mu^1 \geq r$, $\mu_1^0 > r + \mu^1$, and

$$R > (r + \lambda + \mu^1) \frac{\mu_1^0 - r}{\mu_1^0 - r - \mu^1};$$

- (iv) immediate patenting is optimal if $\mu^1 < r$,

$$\mu_1^0 > \mu^1 + \frac{(r + 2\lambda + \mu^1)\mu^1}{2(r + \lambda)},$$

and

$$R > (r + \lambda + \mu^1) \frac{2(\mu_1^0 - \mu^1)(r + \lambda)}{2(\mu_1^0 - \mu^1)(r + \lambda) - (r + 2\lambda + \mu^1)\mu^1};$$

- (v) $1 \parallel 2$ is optimal in all other cases.

Figure 6 shows that the innovator should only patent the innovation before initiating activity 1 if the profit potential R and the imitation rate μ_1^0 are both sufficiently high. Depending on whether

the circumvention rate μ^1 is larger or smaller than the discount rate r , this takes the form of either starting the innocent task first ($2 \rightarrow 1$) or patenting the innovation immediately. If $\mu_1^0 \leq \mu^1$, then it is clearly optimal not to patent the innovation since secrecy offers better protection than patenting. Even when $\mu_1^0 > \mu^1$ it might still be optimal to initiate activity 1 without patenting because if the activity does not cause imitation, then activity 2 can be safely executed since $\mu_2^0 = 0$.

Somewhat unexpectedly, it is sometimes optimal to schedule the suspicious activity first (i.e., $1 \rightarrow 2$), which conflicts with the intuition that it is better to postpone suspicious tasks [10,11]. By scheduling the suspicious task before the innocent one, however, the innovator can test whether imitation occurs without incurring the cost of activity 2. This is similar to scheduling with activity failures, where it can also be optimal to start with those tasks that are most likely to cause a failure [55].

Inspired by Pinker et al. [10], we define the exposed time of a project as the time between the moment of patenting (when the project's existence becomes public knowledge) and the moment of product launch. For a parallel project structure, we obtain that, if no circumvention occurs, immediate patenting and the subsequent execution of activities 1 and 2 in parallel leads to an expected lead time and exposed time both equal to $3/(2\lambda)$. For the hybrid secrecy/patenting strategy $2 \rightarrow 1$, in turn, we have that, if no circumvention occurs, the expected lead time equals $2/\lambda$, whereas the expected exposed time is equal to $1/\lambda$. Therefore, the choice between immediate patenting and $2 \rightarrow 1$ reflects a trade-off between lead time and exposed time. These findings are consistent with the hypothesis of [18] that firms in time-critical industries should rely less on secrecy as it can increase the lead time, as well as with the claim of [14] that secrecy can decrease the exposed time.

5.2.2. Numerical experiments for general project networks.

We now examine the effect of secrecy on the project's lead time in more detail using numerical experiments for general project networks. For all baseline test instances described in Appendix A with $n \in \{11, 21, 31, 41\}$, we compute the expected lead time conditional on no imitation or circumvention occurring (such that there is no project abandonment). We do this for both an optimal and an early-start policy, where this latter policy is defined such that it minimizes the lead time. In particular, it immediately patents the innovation (since $\lambda_i^0 \leq \lambda_i^1$ for each $i \in N$) and subsequently initiates each activity as soon as all its predecessors are completed. We also constructed a second set of test instances based on our original test instances by setting $\lambda_i^1 = \lambda_i^0$ for each $i \in N$. By disregarding the duration benefit from patenting, we can single out the effect of secrecy on the lead time caused by the scheduling decisions.

Figure 7 plots the average percentage increase in the expected lead time under an optimal policy compared to the early-start policy. We see that secrecy, whether or not combined with patenting in a hybrid strategy, can cause a significant increase in the expected lead time. In the case of immediate patenting, in contrast, the expected lead time is, on average, only marginally larger than the one under the early-start policy. Not surprisingly, the major part of this increase in the expected lead time due to secrecy is explained by the duration parameters that differ before and after patenting. Nevertheless, even if we consider instances with no duration benefit from patenting, the expected lead time under a hybrid secrecy/patenting strategy exceeds the one under the early-start policy to a much greater extent than in the case of immediate patenting. This can be explained by the fact that, consistent with our above findings for the parallel project instances, the innovator has an incentive to postpone the execution of certain activities in order to benefit from both the advantages of secrecy as well as patenting.

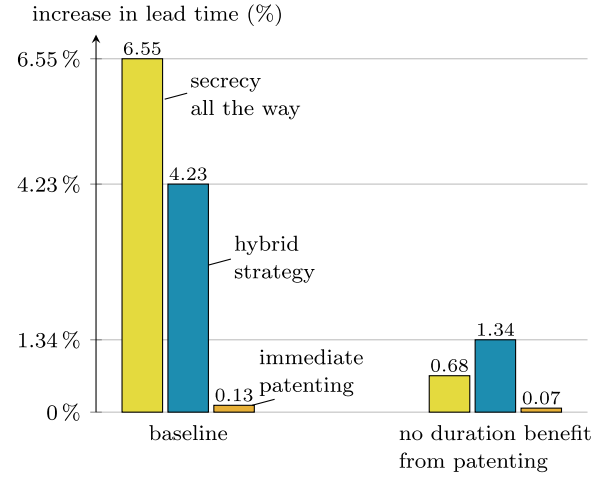


Fig. 7. Percentage increase in the expected lead time under an optimal policy (split down according to its structure) compared to the early-start policy, for general project networks.

6. Discussion

We have introduced a novel model that allows us to examine how a company should manage secrecy during the development of a new product. We find that, for a wide range of parameter settings, the innovator benefits from a hybrid strategy in which both secrecy and patenting are both used in different parts of the project. Hence, contrary to the common view that secrecy and patenting are mutually exclusive, we find that both protection mechanisms can in fact be complementary and that joint deployment regularly yields higher value. A hybrid secrecy/patenting strategy is more likely to be optimal if secrecy becomes more difficult to maintain as the project proceeds. This situation frequently occurs in practice because, at later stages of the project, it typically becomes harder to keep the innovation secret as more people become involved and external input might be required. Simultaneously, our model also indicates when the innovator can rely on simple strategies such as patenting the project immediately or keeping it secret throughout its entire execution. Indeed, in settings where secrecy becomes easier to maintain as the project proceeds, such simple strategies are more likely to be optimal.

With respect to lead-time advantages, which form the third major appropriation method, we find that secrecy has a mixed effect: while it reduces the project's exposed time, it tends to increase its total lead time. In a hybrid secrecy/patenting strategy, this increased lead time results not only from the fact that secrecy can affect efficiency but also from the incentive to postpone the execution of easily-imitable tasks until after patenting, albeit to a lesser extent. Therefore, firms operating in time-critical industries should only rely on secrecy if it affects the project's lead time only to a small extent or if patents are easily circumvented. Somewhat surprisingly, our model shows that, if the profit potential of the project is relatively low, then it might be optimal to initiate first (rather than to postpone) those tasks that are most likely to trigger imitation. By doing so, the innovator keeps sufficient flexibility to abandon the project if needed before the expenses of other activities have been incurred.

Our model also supports policy makers in quantifying how changes in the patenting regime can affect the level of innovation or dissemination of knowledge. We find, for example, that decreasing the patent life may hinder the dissemination of knowledge as it provides an incentive to postpone the moment of patenting. This effect, however, is only marginal if, as is the case in numerous in-

dustries, the expected product life is considerably shorter than the patent life. Our model also indicates that, if the protection provided by patents and secrets is weak, then this might reduce the overall level of innovation because project abandonment would occur more frequently.

The Markov decision process used to formulate the problem is sufficiently flexible to allow for numerous generalizations. The assumption that all activity durations are exponentially distributed, for example, can be generalized to phase-type distributions, which in turn can be used to approximate arbitrary distributions [34,56]. Our assumption that the innovator receives a constant cash inflow during the product's life can also be generalized to more general value profiles by subdividing the final activity into multiple tasks that reflect, for instance, the growth and decline phase in the product life cycle [58]. To model a more general discovery process, in turn, we can allow the imitation rate to depend on both the set of completed activities and on the set of activities that are currently being executed. Moreover, by increasing the number of possible values for the indicator δ in the state definition, we can also incorporate a more gradual discovery such as the one considered by [10]. Finally, by only considering patenting as a feasible action when some activities $P' \subseteq N$ have been completed, one can also incorporate the requirement that certain activities need to be completed before there is sufficient novelty to make a patent defensible. Such generalizations, however, come at the cost of an increase in the model's complexity as well as the computation time required to determine an optimal policy.

Although our Markov decision process is very flexible and can solve realistically-sized instances, it does suffer from the curse of dimensionality. One way to deal with this is to re-execute our stochastic dynamic program multiple times in a rolling-horizon fashion to extract an optimal action for a specified number of next states. Such an approach has as an additional practical benefit that, when re-executing the procedure, one can incorporate all new information that has become available throughout the project's execution so far. Another potential avenue to pursue is the development of heuristic optimization procedures for the proposed optimization problem. Given our formulation as a Markov decision process, a particularly promising approach is to rely on the techniques of approximate dynamic programming [59]. Finally, a third approach to handle the curse of dimensionality is to follow a layered approach in which, initially, different activities are grouped together into work packages, and the more granular scheduling of the activities within a work package is postponed to a lower level of decision-making [60,61].

The setting of managing secrecy during the NPD process provides multiple avenues for further research. In this paper, we have considered a simplified patenting system in which the firm only files for a single patent, which is immediately published at the moment of filing. Although this assumption is common in the literature [3], it would be interesting to generalize our model to incorporate the time lag between filing and publishing (which extends the period of secrecy), as well as the possibility of patenting only a specific fraction or to patent subsequent parts of the innovation [7,21,22]. One convenient way to include the time lag between filing and publishing a patent would be to model it as an optional project activity whose duration corresponds to this time lag.

Another important question is how to combine secrecy with the seemingly opposite trend of open innovation, which has grown to a prominent stream of research over the last decades [1,62]. Although there is an obvious conflict with secrecy in many ways, there can be complementarities as well. In particular, a modular approach to the innovation process makes it possible to shield off the part of the innovation that the company wants to keep secret, while supporting cooperation in other components [63,64]. The ensuing question of which parts to shield off and when to cooperate

highlights the need for a framework that advises managers on how to deal with secrecy during the development process.

Finally, a significant research potential exists in testing the validity of our model empirically. By mapping different industry and firm characteristics to the parameters of our model, one could compare the model's recommendations to (a measure of) firms' actual strategy. To perform this validation, however, detailed and reliable data on the level of the project is needed, which creates a major challenge by the very nature of secrecy.

Author Statement

We certify that the submission is original work by its authors (Ben Hermans, Roel Leus, and Bart Van Looy) and that the work has neither been copyrighted, nor been accepted for publication in another journal, nor been submitted for publication to another journal. The authors declare that they have no relevant or material financial interests that relate to the research described in our paper. All authors contributed equally to the paper.

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Data Availability

No data was used for the research described in the article.

Appendix A. Description of test instances

For all test instances, the project network (N, A) as well as the parameters λ_i^0 and c_i^0 for each $i \in N \setminus \{n\}$ are taken from Creemers et al. [35]. To generate the cost parameters after patenting, we first randomly decide with a 50% probability whether a development task $i \in N \setminus \{n\}$ is more efficient when executed after patenting. If so, then a random number p is drawn from the interval $[0.75, 1]$, and we set $c_i^1 = pc_i^0$. A similar procedure (but with new random numbers) is used to generate the duration parameters after patenting, where we set $p\lambda_i^1 = \lambda_i^0$. Let $q := \sum_{i \in N \setminus \{n\}} 1/\lambda_i^1$ be the expected lead time when the project is patented immediately and when all activities are executed sequentially. We consider a discount rate $r = 0.05$ and, for product launch, we set $1/\lambda_n = 0.5q$ and $\alpha = 0.10$. The end-of-project payoff rate c_n is chosen such that the expected payoff $c_n/(r + \lambda_n)$ equals 100 times the total expected development costs $\sum_{i \in N \setminus \{n\}} |c_i^0|/(r + \lambda_i^0)$ in secrecy. The patent life is set equal to $\ell = 6q$.

The imitation and circumvention rates are generated in such a way that allows us to parameterize on the probability that no imitation or circumvention occurs. To do so, we use as a reference a serial project in which all activity durations are exponentially distributed with an equal parameter $\bar{\lambda}^\delta := \sum_{i \in N \setminus \{n\}} \lambda_i^\delta / (n - 1)$, for $\delta \in \{0, 1\}$. If the time until an event such as imitation or circumvention is exponentially distributed with parameter $\mu^\delta \geq 0$, then the probability of no event before launch equals $(\bar{\lambda}^\delta / (\bar{\lambda}^\delta + \mu^\delta))^{n-1}$. To generate the circumvention rate, we set μ^1 such that this probability equals 70%. To generate the imitation rate, in turn, we set an average rate $\bar{\mu}^0$ such that the aforementioned probability equals 80% if all development tasks have imitation rate $\bar{\mu}^0$. Next, we randomly decide with a 50% probability whether a development task $i \in N \setminus \{n\}$ is suspicious (i.e., has positive μ_i^0), and subsequently set $\mu_i^0 = \bar{\mu}^0$ if i is suspicious and $\mu_i^0 = 0$ otherwise. Finally, the parameter μ_n^0 is chosen such that, at the moment of product launch, the innovator is indifferent between secrecy and patenting.

Table A.1

Considered experimental settings for numerical experiments.

Setting	Modification compared to the baseline setting
increasing imitation rate throughout project	the probability that activity $i \in N \setminus \{n\}$ is suspicious equals i/n
increasing cost difference throughout project	the probability that activity $i \in N \setminus \{n\}$ costs less when executed after patenting equals i/n
increasing duration difference throughout project	the probability that activity $i \in N \setminus \{n\}$ is faster when executed after patenting equals i/n
shorter patent life	$\ell = 0.5 \sum_{i \in N \setminus \{n\}} 1/\lambda_i^1$

We refer to the instances described above as the baseline test instances. For the numerical experiments of Section 5, we also generated additional sets of test instances by changing specific parameters of the baseline setting. Table A.1 describes for each of the considered experimental settings how the generation process was modified to obtain a new set of test instances. For the first three settings, we use the fact that all task indices are such that for each $(i, j) \in A$ it holds that $i < j$ – that is to say, the indices are topologically ordered. Hence, if activity $i \in N \setminus \{n\}$ has a certain property with probability i/n , then we know that, on average, later-occurring activities have a higher probability of having the property.

Appendix B. Estimate for missed profit because of a finite patent life

If the innovator files for a patent when the tasks in $P \in \mathcal{P}$ have been completed, then we assume this immediately generates the following cost:

$$\gamma(P) := e^{-\ell\mu^1} \cdot e^{-(\ell-T(P))\lambda_n} \cdot e^{-r\ell} \cdot \frac{(1-\alpha)c_n}{r + \mu^1 + \lambda_n^1}. \quad (\text{B.1})$$

Here, $T(P)$ is the expected time until product launch, conditional on the information that no patent circumvention occurs. The expression combines (i) the probability that no patent circumvention occurs before product launch; (ii) a lower bound (by Jensen's inequality) for the probability that, after launch, the product outlives the remaining patent life; (iii) the discount rate from patent expiry to the current state; and (iv) the additional expected discounted payoff with an unlimited patent life. The estimate $\gamma(P)$ could deviate from the patent life's true effect because it is based on the expected time until product launch $T(P)$ at the moment of patenting instead of the underlying random variable, which softens the incentive to expedite the project to extend the effective monopoly period. The estimate also does not take into account that the patent might expire before project completion. This latter effect seems rather innocent since, in practice, development times only rarely exceed the typical patent life of 20 years.

To obtain $T(P)$ in a given state $(P, 1) \in \mathcal{S}$ with $P \subset N \setminus \{n\}$, we adapt the recursion of [36]. With $\varphi^*(P, 1)$ as defined in Section 4.2, denote $\lambda^*(P) := \sum_{i \in \varphi^*(P, 1)} \lambda_i^1$. The transition time, conditional on the information that no circumvention occurs, is then exponentially distributed with expected value $1/\lambda^*(P)$. It then holds that

$$T(P) = \frac{1}{\lambda^*(P)} \left(1 + \sum_{i \in \varphi^*(P, 1)} \lambda_i^1 T(P_i) \right).$$

and combined with the boundary condition $T(N \setminus \{n\}) = 0$ this enables us to compute $T(P)$ by a backward recursion based on the cardinality of P .

In order to assess the quality of our estimate, we consider a Monte Carlo simulation for the missed profit due to the finite patent life when patenting the project immediately and following an optimal policy afterwards. This quantity is then compared to $\gamma(\emptyset)$; i.e., our estimate for the missed profit when patenting the project at its initiation. These simulations are run for all instances described in Appendix A with $n \in \{11, 21, 31, 41\}$ under the 'shorter patent life'-setting, since for these instances the missed profit due to the finite patent life was most substantial. Only instances with a positive eNPV were used, because otherwise the project is abandoned immediately and the missed profit due to the finite patent life never materializes. For each of these instances, we computed the average missed profit over 100,000 simulations.

Table B.2 displays the result of this simulation study by reporting on the average absolute deviation between the simulated missed profit versus our estimate $\gamma(\emptyset)$ for this quantity. Table B.2(a) shows this absolute deviation relative to our estimate $\gamma(\emptyset)$ and Table B.2(b) shows this value relative to the maximum eNPV. These values are averaged over all instances for the corresponding value of n and OS with a positive eNPV.

We observe that our estimate for the missed profit due to a finite patent life sometimes deviates substantially from the simulated value. This is mainly caused by the fact that Jensen's inequality only provides a rather weak bound on the probability that after launch the product outlives the remaining patent life (see point (ii) subsequent to Eq. (B.1)). The large deviation for $n = 30$ and OS = 0.80 in Table B.2(b) can be explained by the fact that the maximum eNPV is very close to zero for some of these instances.

The deviation is especially large for instances with a small number of activities and a larger order strength. As reported in Section 4.3, these instances are also exactly those that are the easiest to solve. Hence, in practical settings where the finite patent life seems to have a decisive impact, it is computationally feasible for companies to complement the analytical method described above with a Monte Carlo simulation. For instances with a large number of activities and/or a small order strength, on the other hand, the deviation is much more limited. Based on the values reported in Table B.2(b), it seems unlikely that for these instances the estimation error will affect the optimal policy. As such, for these instances that are computationally more demanding to solve, it seems safe to use the analytical estimate $\gamma(P)$ as described above.

Table B.1Absolute deviation between the simulated average missed profit due to a finite patent life versus our estimate $\gamma(\emptyset)$ for this quantity relative to our estimate (panel a) or the maximum eNPV (panel b).

n	OS = 0.80	OS = 0.60	OS = 0.40	n	OS = 0.80	OS = 0.60	OS = 0.40
11	0.701001	0.498416	0.370209	11	0.057226	0.017840	0.019497
21	0.543914	0.259266	0.139646	21	0.073349	0.013434	0.003092
31	0.381344	0.110275	0.031539	31	1.380151	0.004472	0.000231
41	0.174881	0.032007	0.037843	41	0.110580	0.000877	0.000088

(a) Absolute deviation relative to $\gamma(\emptyset)$.

(b) Absolute deviation relative to maximum eNPV.

Appendix C. Proof of Theorem 1

Consider a non-terminal state $\sigma = (P, \delta) \in \mathcal{S}$. Using Eq. (1) and the definition of $\psi_i(\sigma)$ in Eq. (3), we have for every $\varphi \subseteq E(P)$ that

$$W(P, \delta, \varphi) = \frac{\mu^\delta V(P, 2) + \sum_{j \in \varphi} (\lambda_j^\delta + \mu_j^\delta) \psi_j(\sigma)}{r + \mu^\delta + \sum_{j \in \varphi} (\lambda_j^\delta + \mu_j^\delta)}. \quad (\text{C.1})$$

To establish Theorem 1, we will use the following two lemmas.

Lemma 1. For every $a, b, v, \lambda \in \mathbb{R}$ with $b > 0$ and $\lambda > 0$ it holds that

$$v \geq \frac{a}{b} \text{ if and only if } v \geq \frac{a + \lambda v}{b + \lambda}$$

and

$$v \geq \frac{a}{b} \text{ if and only if } \frac{a + \lambda v}{b + \lambda} \geq \frac{a}{b}.$$

Proof. By rearranging terms and using the fact that $b > 0$, we obtain that

$$v \geq \frac{a + \lambda v}{b + \lambda}$$

is equivalent to $(b + \lambda)v \geq a + \lambda v$, which in turn is equivalent to $v \geq a/b$. Similarly, since $\lambda > 0$,

$$\frac{a + \lambda v}{b + \lambda} \geq \frac{a}{b}$$

is equivalent to $(a + \lambda v)b \geq (b + \lambda)a$, which in turn is equivalent to $v \geq a/b$. $\square$

Lemma 2. For every non-terminal state $\sigma = (P, \delta) \in \mathcal{S}$, eligible task $i \in E(P)$, and action $\varphi \subseteq E(P) \setminus \{i\}$ it holds that $\psi_i(\sigma) \geq W(P, \delta, \varphi)$ if and only if $W(P, \delta, \varphi \cup \{i\}) \geq W(P, \delta, \varphi)$.

Proof. Consider an arbitrary non-terminal state $\sigma = (P, \delta) \in \mathcal{S}$, eligible task $i \in E(P)$, and action $\varphi \subseteq E(P) \setminus \{i\}$. To simplify notation, let $a := \mu^\delta V(P, 2) + \sum_{j \in \varphi} (\lambda_j^\delta + \mu_j^\delta) \psi_j(\sigma)$ and $b := r + \mu^\delta + \sum_{j \in \varphi} (\lambda_j^\delta + \mu_j^\delta)$ such that, by Eq. (C.1), $W(P, \delta, \varphi) = a/b$. Moreover, let $v := \psi_i(\sigma)$ and $\lambda := \lambda_i^\delta + \mu_i^\delta$ such that, by Eq. (C.1),

$$W(P, \delta, \varphi \cup \{i\}) = \frac{a + \lambda v}{b + \lambda}.$$

By using the fact that $b > 0$ and $\lambda > 0$, the result then follows from the second equivalence in Lemma 1 with a, b, v, λ as defined above. $\square$

Observe that if patenting is not optimal in state σ , then $V(\sigma)$ equals the maximum of $W(P, \delta, \varphi)$ over all $\varphi \subseteq E(P)$. Lemma 2 thus implies that executing an activity $i \in E(P)$ with $\psi_i(\sigma) \geq V(\sigma)$ never decreases the eNPV, whereas executing an activity $i \in E(P)$ with $\psi_i(\sigma) < V(\sigma)$ cannot be part of an optimal action because it would decrease the eNPV. Hence, executing the set of activities $\varphi^*(\sigma) = \{i \in E(P) : \psi_i(\sigma) \geq V(\sigma)\}$ forms an optimal action, which proves Theorem 1. $\square$

Appendix D. Proof of Theorem 2

Consider a non-terminal state $\sigma = (P, \delta) \in \mathcal{S}$ where it is not optimal to patent the innovation. Define $I := \{i \in E(P) : \psi_i(\sigma) \geq W(P, \delta, \varphi_i(\sigma))\}$, then we want to show that $I = \varphi^*(\sigma)$. Consider an $i \in E(P)$ with $\psi_i(\sigma) \geq V(\sigma)$, then it follows from Eq. (2) that $\psi_i(\sigma) \geq V(\sigma) \geq W(P, \delta, \varphi_i(\sigma))$. Hence, $\varphi^*(\sigma) \subseteq I$.

To establish $I \subseteq \varphi^*(\sigma)$, we show that $\psi_i(\sigma) \geq V(\sigma)$ for every $i \in I$. Let $j \in I$ be such that $\psi_j(\sigma) = \max_{i \in I} \psi_i(\sigma)$ and thus $\varphi_j(\sigma) = \emptyset$. Suppose that, contrary to what we want to show, $\psi_j(\sigma) < V(\sigma)$. By definition of j , it then follows that $\psi_i(\sigma) < V(\sigma)$ for all $i \in$

I . Since $\varphi^*(\sigma) \subseteq I$, this implies $\varphi^*(\sigma) = \emptyset$ and Theorem 1 yields the contradiction $0 = W(P, \delta, \varphi_j(\sigma)) \leq \psi_j(\sigma) < V(\sigma) = 0$. Hence, $\psi_j(\sigma) \geq V(\sigma)$. Next, define for an arbitrary $k \in I$ the set $J = \{i \in I : \psi_i(\sigma) > \psi_k(\sigma)\}$ and, as the induction hypothesis, assume that $\psi_i(\sigma) \geq V(\sigma)$ for all $i \in J$. Since $\psi_i(\sigma) < V(\sigma)$ for all $i \in E(P) \setminus I$, the induction hypothesis implies that $J = \varphi_k(\sigma) = \{i \in E(P) : \psi_i(\sigma) > \psi_k(\sigma)\}$. Suppose that, contrary to what we want to show, $\psi_k(\sigma) < V(\sigma)$, then it follows that $\varphi^*(\sigma) = \varphi_k(\sigma)$ and, combined with Theorem 1, we obtain the contradiction $\psi_k(\sigma) < V(\sigma) = W(P, \delta, \varphi^*(\sigma)) = W(P, \delta, \varphi_k(\sigma)) \leq \psi_k(\sigma)$. Hence, $\psi_k(\sigma) \geq V(\sigma)$, which concludes the induction step and therefore also the proof. $\square$

Appendix E. Proof of Proposition 1

To prove Proposition 1, we use the following lemma, which shows that, for fixed $\delta \in \{0, 1, 2\}$, the value function is non-decreasing in the set P of completed activities. The result directly generalizes Hermans and Leus [9, Lemma 2] by including elements of secrecy and patenting.

Lemma 3. For each non-terminal state $(P, \delta) \in \mathcal{S}$ with $n \notin E(P)$, it holds for every $i \in E(P)$ that $V(P_i, \delta) \geq V(P, \delta)$.

Proof. Consider a non-terminal state $P \in \mathcal{P}$ with $n \notin E(P)$ and take arbitrary $i \in E(P)$. Referring to the formulas in Appendix B, one can show that $\gamma(P) \geq \gamma(P_i)$. If $V(P, 1) \leq V(P_i, 1)$, we thus obtain that $V(P, 1) - \gamma(P) \leq V(P_i, 1) - \gamma(P_i) \leq V(P_i, 0)$, where the final inequality follows from Eq. (2). Hence, to establish the lemma, it suffices to show that

$$\max_{\varphi \subseteq E(P)} W(P, \delta, \varphi) \leq V(P_i, \delta)$$

for all $\delta \in \{0, 1, 2\}$. Also observe that, since c_n is the unique positive cash flow and $\alpha \in [0, 1]$, the parameters are such that, in an optimal policy, the innovator never profits from imitation, circumvention, or patent expiry. Hence, $V(P, \delta) \geq V(P, 2)$ and $V(P_i, \delta) \geq V(P_i, 2)$ for $\delta \in \{0, 1\}$.

We now prove the lemma by induction on the number of completed tasks $|P|$. Suppose first that $|P| = n - 2$, such that $P = N \setminus \{i, n\}$, $E(P) = \{i\}$, $P_i = N \setminus \{n\}$, and $E(P_i) = \{n\}$. Equations (1)–(2) with $\delta = 2$ then yield that

$$V(P_i, 2) = \max\{0, W(P_i, 2, \{n\})\} = \frac{\alpha c_n}{r + \lambda_n^2}$$

since $c_n > 0$ and $\alpha \in [0, 1]$, and that

$$\begin{aligned} \max\{0, W(P, 2, \{i\})\} &= \max\left\{0, \frac{c_i^2 + \lambda_i^2 V(P_i, 2)}{r + \lambda_i^2}\right\} \\ &\leq \frac{\lambda_i^2}{r + \lambda_i^2} V(P_i, 2) \\ &\leq V(P_i, 2) \end{aligned}$$

since $c_i^2 \leq 0$ and $V(P_i, 2) \geq 0$. Similarly, we obtain for $\delta \in \{0, 1\}$ that

$$\begin{aligned} V(P_i, \delta) &\geq W(P_i, \delta, \{n\}) = \frac{c_n + \mu^\delta(\{n\})V(P_i, 2)}{r + \lambda_n^\delta + \mu^\delta(\{n\})} \\ &\geq \frac{\mu^\delta(\{n\})}{r + \lambda_n^\delta + \mu^\delta(\{n\})} V(P_i, 2) \geq 0 \end{aligned}$$

since $c_n \geq 0$ and $V(P_i, 2) \geq 0$, and that

$$\begin{aligned} W(P, \delta, \{i\}) &= \frac{c_i^\delta + \lambda_i^\delta V(P_i, \delta) + \mu^\delta(\{i\})V(P, 2)}{r + \lambda_i^\delta + \mu^\delta(\{i\})} \\ &\leq \frac{\lambda_i^\delta + \mu^\delta(\{i\})}{r + \lambda_i^\delta + \mu^\delta(\{i\})} V(P_i, \delta) \leq V(P_i, \delta) \end{aligned}$$

since $c_i^\delta \leq 0$, $V(P_i, \delta) \geq 0$, and, as derived above, $V(P, 2) \leq V(P_i, 2) \leq V(P_i, \delta)$. We therefore obtain that $\max\{0, W(P, \delta, \{i\})\} \leq V(P_i, \delta)$ for $\delta \in \{0, 1\}$. Hence, we have established that $V(P_i, \delta) \geq V(P, \delta)$ for all $\delta \in \{0, 1, 2\}$.

Now take arbitrary $P \in \mathcal{P}$ with $|P| < n - 2$ and an $i \in E(P)$. As the induction hypothesis, assume that $V(P'_j, \delta) \geq V(P', \delta)$ for each $P' \in \mathcal{P}$ with $|P'| = |P| + 1$, for each $j \in E(P')$, and for each $\delta \in \{0, 1, 2\}$. For every $\varphi \subseteq E(P)$, define $I(i, \varphi) := 1$ if $i \in \varphi$ and $I(i, \varphi) := 0$ otherwise. Additionally, let $P_{ij} := P \cup \{i, j\}$ for every $j \in E(P) \setminus \{i\}$. Equation (1) then yields that for every $\varphi \subseteq E(P)$

$$\begin{aligned} W(P, 2, \varphi) &\leq \frac{\lambda_i^2 V(P_i, 2) I(i, \varphi) + \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^2 V(P_j, 2) + c_j^2)}{\lambda_i^2 I(i, \varphi) + r + \sum_{j \in \varphi \setminus \{i\}} \lambda_j^2} \\ &\leq \frac{\lambda_i^2 V(P_i, 2) I(i, \varphi) + \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^2 V(P_{ij}, 2) + c_j^2)}{\lambda_i^2 I(i, \varphi) + r + \sum_{j \in \varphi \setminus \{i\}} \lambda_j^2} \leq V(P_i, 2). \end{aligned} \quad (\text{E.1})$$

Here, the first inequality follows from $c_i^2 \leq 0$, and the second from the induction hypothesis. To see the third inequality, observe that $\varphi \setminus \{i\} \subseteq E(P_i)$ such that Eqs. (1)-(2) imply

$$\frac{\sum_{j \in \varphi \setminus \{i\}} (\lambda_j^2 V(P_{ij}, 2) + c_j^2)}{r + \sum_{j \in \varphi \setminus \{i\}} \lambda_j^2} \leq V(P_i, 2).$$

If $I(i, \varphi) = 0$, then this immediately yields Inequality (E.1). Otherwise, define

$$a := \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^2 V(P_{ij}, 2) + c_j^2)$$

$$b := r + \sum_{j \in \varphi \setminus \{i\}} \lambda_j^2,$$

as well as $v := V(P_i, 2)$ and $\lambda := \lambda_i^2 I(i, \varphi)$. Since we then have $b > 0$ and $\lambda > 0$, Inequality (E.1) follows from the first equivalence in Lemma 1. We therefore obtain that $\max_{\varphi \subseteq E(P)} W(P, \delta, \varphi) \leq V(P_i, \delta)$ for $\delta = 2$. To see that this is also true for $\delta \in \{0, 1\}$, observe that Eq. (1) yields for every $\delta \in \{0, 1\}$ and $\varphi \subseteq E(P)$ that

$$\begin{aligned} W(P, \delta, \varphi) &\leq \frac{(\lambda_i^\delta V(P_i, \delta) + \mu_i^\delta V(P, 2)) I(i, \varphi) + \mu^\delta V(P, 2) + \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^\delta V(P_j, \delta) + \mu_j^\delta V(P, 2) + c_j^\delta)}{(\lambda_i^\delta + \mu_i^\delta) I(i, \varphi) + r + \mu^\delta + \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^\delta + \mu_j^\delta)} \\ &\leq \frac{(\lambda_i^\delta + \mu_i^\delta) V(P_i, \delta) I(i, \varphi) + \mu^\delta V(P_i, 2) + \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^\delta V(P_{ij}, \delta) + \mu_j^\delta V(P_i, 2) + c_j^\delta)}{(\lambda_i^\delta + \mu_i^\delta) I(i, \varphi) + r + \mu^\delta + \sum_{j \in \varphi \setminus \{i\}} (\lambda_j^\delta + \mu_j^\delta)} \\ &\leq V(P_i, \delta). \end{aligned}$$

Here, the first inequality follows from $c_i^\delta \leq 0$, and the second from the induction hypothesis and the fact that, as derived above, $V(P, 2) \leq V(P_i, 2) \leq V(P_i, \delta)$. Finally, the third inequality follows from an analogous reasoning as the one leading to Inequality (E.1). We thus obtain that for $\delta \in \{0, 1\}$ also, we have $\max_{\varphi \subseteq E(P)} W(P, \delta, \varphi) \leq V(P_i, \delta)$, which completes the proof. $\square$

Lemma 3 implies that if the project is profitable in its initial state (i.e., if $V(\emptyset, 0) > 0$), then it remains to be profitable for every subsequent state $(P, \delta) \in \mathcal{S}$ with $n \notin E(P)$ as long as no imitation or circumvention occurs. This proves Proposition 3 for every state $(P, \delta) \in \mathcal{S}$ with $n \notin E(P)$. To see that the proposition also holds if $P = N \setminus \{n\}$ such that $n \in E(P)$, observe that $V(N \setminus \{n\}, \delta) > 0$ for $\delta \in \{0, 1\}$, and that $V(N \setminus \{n\}, 2) \geq 0$. Hence, for $P = N \setminus \{n\}$ also, it is optimal to consider abandonment only if imitation or circumvention occurs. $\square$

Appendix F. Proof of Proposition 2

For each $i \in N$, denote $[i] := \{1, \dots, i\}$, and let $[0] := \emptyset$. Since we consider a serial project instance, the state space $\mathcal{S}$ then consists of

a state $([i], \delta)$ for every $i \in \{0, \dots, n\}$ and $\delta \in \{0, 1, 2\}$. The proposition assumes that immediate abandonment is not optimal, such that $V(\emptyset, 0) > 0$ and thus by Lemma 3 also that $V([i], 0) > 0$ for every $i \in \{0, \dots, n-1\}$. Now define for every $i \in \{1, \dots, n\}$ the quantity

$$b(i) := V([i-1], 0) - [V([i-1], 1) - \gamma([i-1])].$$

By definition of the value function, $b(i)$ captures the benefit of having the option not to patent the innovation in state $([i-1], 0)$. Hence, for every $i \in \{1, \dots, n\}$, keeping the project secret and initiating activity i is the unique optimal action in state $([i-1], 0)$ if and only if $b(i) > 0$.

To establish the result, it suffices to show that $b(i) > 0$ implies $b(i+1) > 0$ for every $i \in \{1, \dots, n-1\}$ if Conditions (i)-(iii) are satisfied. Indeed, this then yields that if it is optimal to maintain secrecy in the current state, then the conditions imply that it is also optimal to do so in the subsequent state. Patenting and secrecy are then mutually exclusive because it is either optimal to keep the project secret until product launch or to patent the innovation immediately.

Consider an arbitrary $i \in \{1, \dots, n\}$. Referring to Appendix B, we have that $\gamma([i-1]) = 0$ if $\ell = +\infty$. Condition (i) therefore implies that

$$b(i) = V([i-1], 0) - V([i-1], 1). \quad (\text{F.1})$$

By our assumption that $\alpha = 0$, we also have $V([i-1], 2) = 0$. Since maintaining secrecy in the current state $([i-1], 0)$ and patenting in state $([i], 0)$ forms a feasible action, it follows from Section 4's optimality equations that

$$V([i-1], 0) \geq \frac{\lambda_i^0}{r + \lambda_i^0 + \mu_i^0} \left(V([i], 1) + \frac{c_i^0}{\lambda_i^0} \right). \quad (\text{F.2})$$

Moreover, again by the optimality equations,

$$V([i-1], 1) \geq \frac{\lambda_i^1}{r + \lambda_i^1 + \mu^1} \left(V([i], 1) + \frac{c_i^1}{\lambda_i^1} \right),$$

which is equivalent to

$$V([i], 1) \leq \frac{r + \lambda_i^1 + \mu^1}{\lambda_i^1} V([i-1], 1) - \frac{c_i^1}{\lambda_i^1}. \quad (\text{F.3})$$

Now suppose that, contrary to what we want to show, there is a $j \in \{1, \dots, n-1\}$ such that $b(j) > 0$ and $b(j+1) = 0$. Since $b(j+1) = 0$, we have by Eq. (F.1) that $V([j], 1) = V([j], 0)$. Together with $V([j-1], 0) > 0$, we therefore know that Inequality (F.2) for $i = j$ is met with equality. Combined with Inequality (F.3) for $i = j$, and our assumption that $b(j) > 0$, we obtain

$$\begin{aligned} 0 < b(j) &= V([j-1], 0) - V([j-1], 1) \\ &\leq \frac{\lambda_j^0}{r + \lambda_j^0 + \mu_j^0} \left(\frac{r + \lambda_j^1 + \mu^1}{\lambda_j^1} V([j-1], 1) - \frac{c_j^1}{\lambda_j^1} + \frac{c_j^0}{\lambda_j^0} \right) - V([j-1], 1). \end{aligned}$$

After multiplying both sides of the inequality with $(r + \lambda_j^0 + \mu_j^0)/\lambda_j^0$ and rearranging terms, this yields

$$0 < \left(\frac{r + \mu^1}{\lambda_j^1} - \frac{r + \mu_j^0}{\lambda_j^0} \right) V([j-1], 1) - \frac{c_j^1}{\lambda_j^1} + \frac{c_j^0}{\lambda_j^0}. \quad (\text{F.4})$$

Since $V([j-1], 1) \geq 0$ and $c_j^0/\lambda_j^0 - c_j^1/\lambda_j^1 \leq 0$, it therefore must be the case that

$$\frac{r + \mu^1}{\lambda_j^1} - \frac{r + \mu_j^0}{\lambda_j^0} > 0.$$

Inequality (F.4), Conditions (ii)-(iii), and Lemma 3 therefore yield

$$0 < \left(\frac{r + \mu^1}{\lambda_{j+1}^1} - \frac{r + \mu_{j+1}^0}{\lambda_{j+1}^0} \right) V([j], 1) + \frac{c_{j+1}^0}{\lambda_{j+1}^0} - \frac{c_{j+1}^1}{\lambda_{j+1}^1}. \quad (\text{F.5})$$

On the other hand, $V([j], 1) > 0$ since $V([j], 1) = V([j], 0) > 0$, such that Inequality (F.3) for $i = j+1$ is met with equality. Combined with Inequality (F.2) for $i = j+1$, and our assumption that $b(j+1) = 0$, we obtain

$$0 = b(j) = V([j], 0) - V([j], 1) \\ \geq \frac{\lambda_{j+1}^0}{r + \lambda_{j+1}^0 + \mu_{j+1}^0} \left(\frac{r + \lambda_{j+1}^1 + \mu^1}{\lambda_{j+1}^1} V([j], 1) - \frac{c_{j+1}^1}{\lambda_{j+1}^1} + \frac{c_{j+1}^0}{\lambda_{j+1}^0} \right) - V([j], 1),$$

which in turn implies

$$0 \geq \left(\frac{r + \mu^1}{\lambda_{j+1}^1} - \frac{r + \mu_{j+1}^0}{\lambda_{j+1}^0} \right) V([j], 1) + \frac{c_{j+1}^0}{\lambda_{j+1}^0} - \frac{c_{j+1}^1}{\lambda_{j+1}^1}.$$

Comparing this with Inequality (F.5), we have derived a contradiction. Hence, if Conditions (i)-(iii) are satisfied, then $b(i) > 0$ implies $b(i+1) > 0$ for every $i \in \{1, \dots, n-1\}$. This establishes the sufficiency of Conditions (i)-(iii) for patenting and secrecy to be mutually exclusive. $\square$

Appendix G. Proof of Proposition 3

Lemmas 4–9 below characterize when it is optimal to abandon the project, when to initiate activities 1 or 2, and when to patent immediately. Together, they yield Parts (i)-(iv) of Proposition 3, and thus establish the result.

To simplify notation, let $\beta := \min\{\mu_1^0, \mu^1\}$ and

$$\rho := \lambda \frac{c_3}{r + \lambda_3 + \mu^1} + c,$$

such that $R = \rho\lambda/|c|$. We also denote $\rho^+ := \max\{\rho, 0\}$. Recall that we assumed $\ell = +\infty$ and $\alpha = 0$ such that $\gamma(P) = 0$ and $V(P, 2) = 0$ for all $P \in \mathcal{P}$. It then immediately follows from Section 4's optimality equations that

$$V(\{1\}, 1) = V(\{2\}, 1) = \frac{\rho^+}{r + \lambda + \mu^1}, \quad (\text{G.1})$$

$$V(\{1\}, 0) = \max \left\{ \frac{\rho^+}{r + \lambda}, \frac{\rho^+}{r + \lambda + \mu^1} \right\} = \frac{\rho^+}{r + \lambda}, \text{ and} \quad (\text{G.2})$$

$$V(\{2\}, 0) = \max \left\{ \frac{\rho^+}{r + \lambda + \mu_1^0}, \frac{\rho^+}{r + \lambda + \mu^1} \right\} = \frac{\rho^+}{r + \lambda + \beta}. \quad (\text{G.3})$$

In particular, if $\rho \leq 0$, then $\rho^+ = 0$, and it is optimal to abandon the project in every state $(\{i\}, \delta)$ with $i \in \{1, 2\}$ and $\delta \in \{0, 1\}$. If $\rho \geq 0$, in turn, it is optimal to initiate the eligible activity in every state $(\{i\}, \delta)$ with $i \in \{1, 2\}$ and $\delta \in \{0, 1\}$, except for state $(\{2\}, 0)$ where it is optimal to patent the innovation first if $\mu^1 < \mu_1^0$.

The following two lemmas characterize when it is optimal to abandon the project in state $(\emptyset, 1)$ and $(\emptyset, 0)$, respectively. In particular, Lemma 5 establishes Part (i) of Proposition 3.

Lemma 4. *In state $(\emptyset, 1)$, it is optimal to abandon the project if and only if $R \leq r + \lambda + \mu^1$, and to initiate both activities 1 and 2 if and only if $R \geq r + \lambda + \mu^1$.*

Proof. Section 4's optimality equations imply that $V(\emptyset, 1) = \max_{\varphi \subseteq \{1,2\}} W(\emptyset, 1, \varphi)$. It therefore follows from Eq. (1) that abandonment is optimal if and only if

$$\lambda V(\{1\}, 1) + c = \lambda V(\{2\}, 1) + c \leq 0.$$

Using Eq. (G.1), this is equivalent to requiring that

$$\frac{\rho^+}{r + \lambda + \mu^1} + \frac{c}{\lambda} \leq 0,$$

which in turn is equivalent to $R = \rho\lambda/|c| \leq r + \lambda + \mu^1$, as required. Analogously, we obtain that initiating both activities 1 and 2 is optimal if and only if $R \geq r + \lambda + \mu^1$. $\square$

Lemma 5. *In state $(\emptyset, 0)$, it is optimal to abandon the project if and only if $R \leq r + \lambda$.*

Proof. Section 4's optimality equations and Eq. (1) yield that $V(\emptyset, 0) = 0$ if and only if

$$\begin{cases} V(\emptyset, 1) = 0; \\ \lambda V(\{1\}, 0) + c \leq 0; \\ \lambda V(\{2\}, 0) + c \leq 0. \end{cases}$$

Since $V(\{2\}, 0) \leq V(\{1\}, 0)$ by Eqs. (G.2)-(G.3), the latter two conditions are both satisfied if and only if

$$\frac{\rho^+}{r + \lambda} + \frac{c}{\lambda} \leq 0,$$

which is equivalent to $R = \rho\lambda/|c| \leq r + \lambda$. By Lemma 4, this also implies that $V(\emptyset, 1) = 0$, which therefore yields the result. $\square$

The following lemma shows that the policies defined in Table 5 suffice to describe an optimal policy for all possible parameter settings.

Lemma 6. *At least one of the policies defined in Table 5 is optimal.*

Proof. Since we know from Lemma 5 that immediate abandonment is optimal if $R \leq r + \lambda$, we can assume that $R > r + \lambda$. Hence, $V(\emptyset, 0) > 0$ and the potentially optimal actions in state $(\emptyset, 0)$ are to execute a set of activities $\varphi \subseteq \{1, 2\}$ with $\varphi \neq \emptyset$ or to patent. It then follows from Lemma 4 and Eqs. (G.1)-(G.3) that choosing an optimal action in state $(\emptyset, 0)$ and continuing with the matching policy described in Table 5 must be optimal. $\square$

The following three lemmas characterize when the policies defined in Table 5 outperform each other. Here, we say that one policy outperforms another if it leads to a strictly higher eNPV. Combined with Lemma 6, they will establish Parts (ii)-(iv) of Proposition 3.

Lemma 7. *If $R > r + \lambda$, then $1 \rightarrow 2$ outperforms $1 \parallel 2$ if and only if $\mu_1^0 \leq \mu^1$ and*

$$R < (r + \lambda) \left(1 + \frac{\mu_1^0}{r} \right),$$

or $\mu_1^0 \geq \mu^1$ and

$$R < (r + \lambda + \mu^1) \frac{(r + \lambda)(r + \mu_1^0)}{(r + \lambda)(r + \mu_1^0) - \lambda\mu^1}. \quad (\text{G.4})$$

Proof. We show the result by characterizing when $1 \rightarrow 2$ does not outperform $1 \parallel 2$. Since $R > r + \lambda$, the policies $1 \rightarrow 2$ and $1 \parallel 2$ lead to an eNPV equal to $W(\emptyset, 0, \{1\})$ and $W(\emptyset, 0, \{1, 2\})$, respectively. Hence, $1 \rightarrow 2$ does not outperform $1 \parallel 2$ if and only if $W(\emptyset, 0, \{1, 2\}) \geq W(\emptyset, 0, \{1\})$. By Lemma 2, this condition is equivalent to $\psi_2(\emptyset, 0) \geq W(\emptyset, 0, \{1\})$.

By definition of $\psi_2(\emptyset, 0)$ in Eq. (3),

$$\begin{aligned}\psi_2(\emptyset, 0) &= \frac{1}{\lambda_2^0 + \mu_2^0} (\lambda_2^0 V(\{2\}, 0) + \mu_2^0 V(\emptyset, 2) + c_2^0) \\ &= V(\{2\}, 0) + \frac{c}{\lambda} = \frac{\rho}{r + \lambda + \beta} + \frac{c}{\lambda},\end{aligned}$$

where the final equality follows from Eq. (G.3) and our assumption that $R > r + \lambda$. Eqs. (1) and (G.2), and our assumption that $R > r + \lambda$ also give

$$W(\emptyset, 0, \{1\}) = \frac{\lambda V(\{1\}, 0) + c}{r + \lambda + \mu_1^0} = \frac{\lambda}{r + \lambda + \mu_1^0} \cdot \frac{\rho}{r + \lambda} + \frac{c}{r + \lambda + \mu_1^0}.$$

Combined, we obtain that $\psi_2(\emptyset, 0) \geq W(\emptyset, 0, \{1\})$ if and only if

$$\begin{aligned}\left(\frac{1}{r + \lambda + \beta} - \frac{\lambda}{(r + \lambda + \mu_1^0)(r + \lambda)} \right) \rho &\geq \frac{|c|}{\lambda} \left(1 - \frac{\lambda}{r + \lambda + \mu_1^0} \right) \\ &= \frac{|c|}{\lambda} \frac{r + \mu_1^0}{r + \lambda + \mu_1^0},\end{aligned}$$

which is equivalent to

$$\frac{(r + \lambda + \mu_1^0)(r + \lambda) - \lambda(r + \lambda + \beta)}{(r + \lambda + \beta)(r + \lambda)} \rho \geq \frac{|c|}{\lambda} (r + \mu_1^0).$$

Now if $\mu_1^0 \leq \mu^1$ such that $\beta = \mu_1^0$, then this yields the condition

$$R = \rho \frac{\lambda}{|c|} \geq \frac{(r + \mu_1^0)(r + \lambda)}{r} = (r + \lambda) \left(1 + \frac{\mu_1^0}{r} \right). \quad (\text{G.5})$$

On the other hand, if $\mu_1^0 \geq \mu^1$ such that $\beta = \mu^1$, then we obtain the condition

$$R = \rho \frac{\lambda}{|c|} \geq (r + \lambda + \mu^1) \frac{(r + \lambda)(r + \mu_1^0)}{(r + \lambda)(r + \mu_1^0) - \lambda \mu^1}. \quad (\text{G.6})$$

Hence, $1 \rightarrow 2$ does not outperform $1 \parallel 2$ if and only if Inequalities (G.5) and (G.6) are satisfied when $\mu_1^0 \leq \mu^1$ and $\mu_1^0 \geq \mu^1$, respectively. $\square$

Lemma 8. If $R > r + \lambda$, then $2 \rightarrow 1$ outperforms $1 \rightarrow 2$ and $1 \parallel 2$ if and only if $\mu_1^0 > r + \mu^1$ and

$$R > (r + \lambda + \mu^1) \frac{\mu_1^0 - r}{\mu_1^0 - r - \mu^1}. \quad (\text{G.7})$$

Proof. We first characterize when $2 \rightarrow 1$ does not outperform $1 \rightarrow 2$. Since $R > r + \lambda$, the policies $2 \rightarrow 1$ and $1 \rightarrow 2$ lead to an eNPV equal to $W(\emptyset, 0, \{2\})$ and $W(\emptyset, 0, \{1\})$, respectively. Hence, $2 \rightarrow 1$ does not outperform $1 \rightarrow 2$ if and only if $W(\emptyset, 0, \{1\}) \geq W(\emptyset, 0, \{2\})$. Eqs. (1) and (G.2)–(G.3) yield that

$$W(\emptyset, 0, \{1\}) = \frac{\lambda V(\{1\}, 0) + c}{r + \lambda + \mu_1^0} = \frac{\lambda}{r + \lambda} \cdot \frac{\rho}{r + \lambda + \mu_1^0} + \frac{c}{r + \lambda + \mu_1^0}$$

and

$$W(\emptyset, 0, \{2\}) = \frac{\lambda V(\{2\}, 0) + c}{r + \lambda} = \frac{\lambda}{r + \lambda} \cdot \frac{\rho}{r + \lambda + \beta} + \frac{c}{r + \lambda}. \quad (\text{G.8})$$

It therefore follows that $W(\emptyset, 0, \{1\}) \geq W(\emptyset, 0, \{2\})$ if and only if

$$(r + \lambda) \left(\frac{\rho}{r + \lambda} + \frac{c}{\lambda} \right) \geq (r + \lambda + \mu_1^0) \left(\frac{\rho}{r + \lambda + \beta} + \frac{c}{\lambda} \right),$$

which is equivalent to

$$\frac{\mu_1^0 - \beta}{r + \lambda + \beta} \rho \leq \mu_1^0 \frac{|c|}{\lambda}.$$

Since $R = \rho \lambda / |c| > r + \lambda$ and $\mu_1^0 \geq 0$, this condition is always satisfied if $\mu_1^0 \leq \mu^1$ such that $\beta = \mu_1^0$. If $\mu_1^0 > \mu^1$, on the other hand, then it is satisfied if and only if

$$R = \rho \frac{\lambda}{|c|} \leq \frac{\mu_1^0}{\mu_1^0 - \mu^1} (r + \lambda + \mu^1).$$

From this, we obtain that $2 \rightarrow 1$ outperforms $1 \rightarrow 2$ if and only if $\mu_1^0 > \mu^1$ and

$$R > (r + \lambda + \mu^1) \frac{\mu_1^0}{\mu_1^0 - \mu^1}. \quad (\text{G.9})$$

Next, we characterize when $2 \rightarrow 1$ outperforms $1 \rightarrow 2$, but does not outperform $1 \parallel 2$. Since $R > r + \lambda$, the policies $2 \rightarrow 1$ and $2 \parallel 1$ lead to an eNPV equal to $W(\emptyset, 0, \{2\})$ and $W(\emptyset, 0, \{1, 2\})$, respectively. Hence, $2 \rightarrow 1$ does not outperform $1 \parallel 2$ if and only if $W(\emptyset, 0, \{1, 2\}) \geq W(\emptyset, 0, \{2\})$, which by Lemma 2 is equivalent to $W(\emptyset, 0, \{2\}) \leq \psi_1(\emptyset, 0)$. By definition of $\psi_1(\emptyset, 0)$ in Eq. (3),

$$\begin{aligned}\psi_1(\emptyset, 0) &= \frac{1}{\lambda_1^0 + \mu_1^0} (\lambda_1^0 V(\{1\}, 0) + \mu_1^0 V(\emptyset, 2) + c_1^0) \\ &= \frac{\lambda V(\{1\}, 0) + c}{\lambda + \mu_1^0} = \frac{\lambda}{\lambda + \mu_1^0} \cdot \frac{\rho}{r + \lambda} + \frac{c}{\lambda + \mu_1^0},\end{aligned}$$

where the final equality follows from Eq. (G.2) and our assumption that $R > r + \lambda$. Combined with Eq. (G.8), and using $\beta = \mu^1$ since $2 \rightarrow 1$ outperforms $1 \rightarrow 2$ and therefore $\mu_1^0 > \mu^1$, we obtain that $\psi_1(\emptyset, 0) \geq W(\emptyset, 0, \{2\})$ if and only if

$$\begin{aligned}\left(\frac{1}{\lambda + \mu_1^0} - \frac{1}{r + \lambda + \mu^1} \right) \rho &\geq \frac{|c|}{\lambda} (r + \lambda) \left(\frac{1}{\lambda + \mu_1^0} - \frac{1}{r + \lambda} \right) \\ &= \frac{|c|}{\lambda} \cdot \frac{r - \mu_1^0}{\lambda + \mu_1^0},\end{aligned}$$

which in turn is equivalent to

$$(r + \mu^1 - \mu_1^0) \rho \geq \frac{|c|}{\lambda} (r - \mu_1^0)(r + \lambda + \mu^1). \quad (\text{G.10})$$

Since $2 \rightarrow 1$ outperforms $1 \rightarrow 2$, Inequality (G.9) is satisfied, and thus $R = \rho \lambda / |c| > (r + \lambda + \mu^1)$. Hence, if $\mu_1^0 \leq r + \mu^1$, then Inequality (G.10) is always satisfied. If $\mu_1^0 > r + \mu^1$, on the other hand, then Inequality (G.10) is equivalent to

$$R = \rho \frac{\lambda}{|c|} \leq (r + \lambda + \mu^1) \frac{\mu_1^0 - r}{\mu_1^0 - r - \mu^1}. \quad (\text{G.11})$$

Finally, observe that $\mu_1^0 > r + \mu^1$ also implies $\mu_1^0 > \mu^1$ and

$$\frac{\mu_1^0 - r}{\mu_1^0 - r - \mu^1} > \frac{\mu_1^0}{\mu_1^0 - \mu^1},$$

such that Inequality (G.9) is satisfied whenever Inequality (G.11) is violated and $\mu_1^0 > r + \mu^1$. From this, we obtain that $2 \rightarrow 1$ outperforms both $1 \rightarrow 2$ and $1 \parallel 2$ if and only if $\mu_1^0 > r + \mu^1$ and Inequality (G.11) is not satisfied, which completes the proof. $\square$

Lemma 9. Immediate patenting outperforms the other policies in Table 5 if and only if $\mu^1 < r$,

$$\mu_1^0 > \mu^1 + \frac{(r + 2\lambda + \mu^1)\mu^1}{2(r + \lambda)}, \quad (\text{G.12})$$

and

$$R > (r + \lambda + \mu^1) \frac{2(\mu_1^0 - \mu^1)(r + \lambda)}{2(\mu_1^0 - \mu^1)(r + \lambda) - (r + 2\lambda + \mu^1)\mu^1}. \quad (\text{G.13})$$

Proof. We first claim that $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$ form necessary conditions for immediate patenting to outperform the other policies. Lemma 4 implies that $R > r + \lambda + \mu^1$ forms a necessary

condition, because otherwise immediate patenting would not outperform immediate abandonment. Now suppose that $R > r + \lambda + \mu^1$ and $\mu^1 \geq \mu_1^0$. Lemma 4, Eq. (G.1), and the facts that $V(\{1\}, 0) \geq V(\{1\}, 1)$ and $V(\{2\}, 0) \geq V(\{2\}, 1)$ then yield

$$\begin{aligned} W(\emptyset, 0, \{1, 2\}) &= \frac{\lambda V(\{1\}, 0) + \lambda V(\{2\}, 0) + 2c}{r + 2\lambda + \mu_1^0} \\ &\geq \frac{\lambda V(\{1\}, 1) + \lambda V(\{2\}, 1) + 2c}{r + 2\lambda + \mu^1} = V(\emptyset, 1), \end{aligned}$$

which shows that immediate patenting does not outperform $1 \parallel 2$. Hence, $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$ are indeed necessary for immediate patenting to outperform the other policies. Now observe that, if $R > r + \lambda + \mu^1$, it then follows from Lemma 4, Eq. (G.1), and Section 4's optimality equations that immediate patenting outperforms the other policies in Table 5 if and only if

$$\begin{aligned} V(\emptyset, 1) &= \frac{\lambda V(\{1\}, 1) + c + \lambda V(\{2\}, 1) + c}{r + 2\lambda + \mu^1} = \frac{2\lambda V(\{2\}, 1) + 2c}{r + 2\lambda + \mu^1} \\ &> \max\{W(\emptyset, 0, \{1\}), W(\emptyset, 0, \{2\}), W(\emptyset, 0, \{1, 2\})\}. \end{aligned} \quad (\text{G.14})$$

To establish the lemma, we proceed in two steps. First, we show that if $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$ hold, then $V(\emptyset, 1) > \max\{W(\emptyset, 0, \{2\}), W(\emptyset, 0, \{1, 2\})\}$ if and only if $\mu^1 < r$ and Inequalities (G.12)–(G.13) are satisfied. Next, we show that Inequalities (G.12)–(G.13) imply the necessary conditions $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$, as well as the inequality $V(\emptyset, 1) > W(\emptyset, 0, \{1\})$. Combined, this then yields the result.

Assume that $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$. We then claim that $V(\emptyset, 1) > W(\emptyset, 0, \{2\})$ if and only if $\mu^1 < r$ and that $V(\emptyset, 1) > W(\emptyset, 0, \{1, 2\})$ if and only if Inequalities (G.12)–(G.13) are satisfied. Since $\mu^1 < \mu_1^0$, we have $\beta = \mu^1$ and thus Eqs. (G.1) and (G.3) imply $V(\{2\}, 1) = V(\{2\}, 0)$. Moreover, Eq. (G.14) holds since $R > r + \lambda + \mu^1$. Hence, $V(\emptyset, 1) > W(\emptyset, 0, \{2\})$ if and only if

$$\begin{aligned} V(\emptyset, 1) &= \frac{2\lambda V(\{2\}, 1) + 2c}{r + 2\lambda + \mu^1} = \frac{\lambda V(\{2\}, 1) + c}{(r + \mu^1)/2 + \lambda} \\ &> \frac{\lambda V(\{2\}, 1) + c}{r + \lambda} = W(\emptyset, 0, \{2\}), \end{aligned}$$

which in turn is equivalent to $\mu^1 < r$. This proves that $V(\emptyset, 1) > W(\emptyset, 0, \{2\})$ if and only if $\mu^1 < r$. Next, observe that $V(\emptyset, 1) > W(\emptyset, 0, \{1, 2\})$ if and only if

$$\begin{aligned} V(\emptyset, 1) &= \frac{2\lambda V(\{2\}, 1) + 2c}{r + 2\lambda + \mu^1} > \frac{\lambda V(\{1\}, 0) + \lambda V(\{2\}, 0) + 2c}{r + 2\lambda + \mu_1^0} \\ &= W(\emptyset, 0, \{1, 2\}), \end{aligned}$$

which, using $V(\{2\}, 0) = V(\{2\}, 1)$ since $\mu^1 < \mu_1^0$, is equivalent to

$$\begin{aligned} &(2(r + 2\lambda + \mu_1^0) - (r + 2\lambda + \mu^1))V(\{2\}, 1) \\ &- (r + 2\lambda + \mu^1)V(\{1\}, 0) > 2\frac{|c|}{\lambda}(r + 2\lambda + \mu_1^0 - r - 2\lambda - \mu^1). \end{aligned}$$

By Eqs. (G.2)–(G.3), this is equivalent to

$$\begin{aligned} &(r + 2\lambda + 2\mu_1^0 - \mu^1)\frac{\rho}{r + \lambda + \mu^1} - (r + 2\lambda + \mu^1)\frac{\rho}{r + \lambda} \\ &> 2\frac{|c|}{\lambda}(\mu_1^0 - \mu^1), \end{aligned}$$

and thus

$$\begin{aligned} &\frac{(r + 2\lambda + 2\mu_1^0 - \mu^1)(r + \lambda) - (r + 2\lambda + \mu^1)(r + \lambda + \mu^1)}{(r + \lambda + \mu^1)(r + \lambda)} R \\ &> 2(\mu_1^0 - \mu^1). \end{aligned} \quad (\text{G.16})$$

Now observe that the numerator of the fraction in the left-hand side of this inequality equals

$$\begin{aligned} &(r + 2\lambda + \mu^1)(r + \lambda) + 2(\mu_1^0 - \mu^1)(r + \lambda) \\ &- (r + 2\lambda + \mu^1)(r + \lambda) - (r + 2\lambda + \mu^1)\mu^1 \\ &= 2(\mu_1^0 - \mu^1)(r + \lambda) - (r + 2\lambda + \mu^1)\mu^1, \end{aligned} \quad (\text{G.17})$$

which is positive if and only if Inequality (G.12) is satisfied. Since $\mu_1^0 > \mu^1$ and $R > r + \lambda + \mu^1 > 0$, we obtain that $V(\emptyset, 1) > W(\emptyset, 0, \{1, 2\})$ only if Inequality (G.12) is satisfied. Rewriting Inequality (G.16) by using Eq. (G.17) and rearranging terms then shows that $V(\emptyset, 1) > W(\emptyset, 0, \{1, 2\})$ if and only if Inequalities (G.12)–(G.13) are satisfied, proving our claim.

It remains to be shown that Inequalities (G.12)–(G.13) imply the necessary conditions $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$, as well as the inequality $V(\emptyset, 1) > W(\emptyset, 0, \{1\})$. Observe that Inequality (G.16) immediately implies $\mu^1 < \mu_1^0$. Moreover, if Inequality (G.12) is satisfied, then Inequality (G.13) is equivalent to

$$R > (r + \lambda + \mu^1) \left(1 - \frac{(r + 2\lambda + \mu^1)\mu^1}{2(\mu_1^0 - \mu^1)(r + \lambda)} \right)^{-1}.$$

We also have that

$$1 > \frac{(r + 2\lambda + \mu^1)\mu^1}{2(\mu_1^0 - \mu^1)(r + \lambda)} > \frac{\lambda\mu^1}{(r + \lambda)(r + \mu_1^0)} \geq 0,$$

where the first inequality follows from Inequality (G.12), and the second one from $(r + 2\lambda + \mu^1)\mu^1 > 2\lambda\mu^1$ and $2(\mu_1^0 - \mu^1) \leq 2\mu_1^0 < 2(r + \mu_1^0)$. As a result, if Inequalities (G.12)–(G.13) are satisfied, then it holds that

$$R > (r + \lambda + \mu^1) \left(1 - \frac{\lambda\mu^1}{(r + \lambda)(r + \mu_1^0)} \right)^{-1} > (r + \lambda + \mu^1).$$

By Lemma 7, the first inequality implies that $1 \rightarrow 2$ does not outperform $1 \parallel 2$. Moreover, since we have derived that immediate patenting outperforms $1 \parallel 2$ if $\mu^1 < \mu_1^0$ and $R > r + \lambda + \mu^1$, we thus obtain that immediate patenting also outperforms $1 \rightarrow 2$ if Inequalities (G.12)–(G.13) are satisfied, which completes the proof. $\square$

As already indicated above, Lemma 5 establishes Part (i) of Proposition 3. We now use Lemmas 6–9 to also establish the remaining parts. In the discussion below, we assume that $R > r + \lambda$ such that immediate abandonment is not optimal.

Observe first that Inequality (G.4) is equivalent to

$$R < (r + \lambda + \mu^1) \left(1 - \frac{\lambda\mu^1}{(r + \lambda)(r + \mu_1^0)} \right)^{-1}.$$

Moreover, if $\mu_1^0 > \mu^1 + r$, then

$$0 \leq \frac{\lambda\mu^1}{(r + \lambda)(r + \mu_1^0)} < \frac{\mu^1}{\mu_1^0 - r} < 1,$$

where the second inequality follows since $\lambda/(r + \lambda) \leq 1$ and $r + \mu_1^0 > \mu_1^0 - r$, and the final one since $\mu_1^0 > \mu^1 + r$. As a result, the conditions derived in Lemma 7 are such that if $1 \rightarrow 2$ outperforms $1 \parallel 2$, then $1 \parallel 2$ outperforms $2 \rightarrow 1$. Additionally, as we have derived in the proof of Lemma 9, if immediate patenting outperforms $1 \parallel 2$, then it also outperforms $1 \rightarrow 2$. Hence, we obtain that if $1 \rightarrow 2$ outperforms $1 \parallel 2$, then it outperforms $2 \rightarrow 1$ and immediate patenting. This establishes Part (ii) of Proposition 3. Moreover, it follows from the proof of Lemma 9 that if immediate patenting outperforms $1 \parallel 2$, then it outperforms $2 \rightarrow 1$ if and only if $\mu^1 < r$. Combined with Lemma 8, this establishes Part (iii) of Proposition 3. Lemma 9, in turn, immediately yields Part (v) of Proposition 3. Finally, from Lemma 6, we have that $1 \parallel 2$ must be optimal for all remaining cases, which then yields Part (iv) and completes the proof of Proposition 3. $\square$

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